

GRADIENT TOXICITY: WHEN AUXILIARY HEADS COLLAPSE THE SHARED ENCODER

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ABSTRACT

Auxiliary objectives are assumed to help a shared encoder. We identify a failure mode in which they destroy it, and which gradient-conflict methods cannot see because the offending gradients are *aligned* rather than opposed: any auxiliary head with a constant attractor — distribution matching, reconstruction, view invariance — contributes a gradient along a single variance-degenerating direction, and above a measurable dose the shared representation collapses. An exact parameter-space identity separates auxiliary contraction from primary restoration. Under explicitly stated scalar assumptions a saddle-node model puts the critical dose at $\alpha_c = 1/\rho$, and a separate spectral analysis distinguishes first-modal contraction from full collapse; for a specified positive semidefinite operator we prove the minimum deletion rank that bounds its compressed spectrum. The mechanism is falsifiable and survives the test: across five domains and two encoders a single healthy probe $\hat{\rho} = 110.2$ predicts $\alpha_c = 9.1 \times 10^{-3}$, inside the independently measured interval $[0.001, 0.01]$, and filtering drives the collapse rate from 10/10 to 0/10 in every domain. Three results delimit it. The primary metric does not measure it — on NYUv2 depth error *improves* under collapse, and on MATR detachment improves SOH-RMSE from 5.20 to 2.71 while also improving observed-RUL error from 118 to 45, so protection and task accuracy are not in tension once right-censoring is handled correctly. Representation PCA need not align with the reconstruction gradient, so a principal-basis surrogate can over-delete. And probe depth alone does not establish spectral optimality. What remains is a diagnostic measurable before deployment, and a filter whose only free parameter is read off one healthy probe.

1 INTRODUCTION

Multi-task learning (MTL) rests on one working assumption: auxiliary tasks supply gradients that steer a shared encoder toward more generalizable representations. PCGrad (Yu et al., 2020), FAMO (Liu et al., 2023), and CAGrad (Liu et al., 2021a) handle the case in which it fails through *conflict* — gradients pushing in different directions — and all presuppose that each task contributes useful directional information.

The assumption can also fail in a way that conflict resolution cannot detect: auxiliary gradients may all point in the *same* direction, and that direction may lead to a degenerate fixed point. `.detach()` is the folk remedy, but it discards the auxiliary signal wholesale: the question is *which* projection to remove, and stop-gradient removes all of them.

We first met this failure in multi-task financial ranking, where a production model with a mixture-density network, an autoencoder, and a contrastive head froze at a near-constant prediction out of sample (+0.0000 rank-IC) that further training never recovered. The failure is not specific to finance. Every auxiliary head of this kind carries a *constant attractor*: distributional, reconstruction, and view-matching objectives are all minimized at constant encoder outputs, so their gradients contract representation variance along the shared direction $u_c = h - \mathbb{E}_b[h]$, regardless of the primary task. We call this *gradient toxicity*, and the resulting *representational collapse*. It is categorically distinct from conflict, because toxic gradients from different heads are *aligned* on a single variance-degenerating axis.

We investigate when auxiliary gradients contract useful representations, which diagnostics reveal the effect, and what filtering can preserve. The theoretical contribution distinguishes an exact signed variance identity, a conditional scalar transition model, and a minimum-rank PSD compression result. Their scopes differ: a practical PCA probe does not automatically inherit a Hessian theorem. Experiments compare full detachment with selective filters, identify basis sensitivity for real reconstruction heads, and examine battery task error alongside representation and prediction variation.

Our contributions are three. First, we identify gradient toxicity as a failure mode orthogonal to conflict — auxiliary gradients that *agree* while contracting a shared representation — and give an exact parameter-space variance identity that separates the contraction from primary restoration, with the assumptions under which scalar and spectral dynamics follow. Second, we characterize minimum-rank deletion for a specified positive semidefinite compression and derive a residual-energy basis criterion for reconstruction gradients. Third, we validate the mechanism across five domains and two encoders, where a one-probe calibration predicts the measured transition and filtering removes the collapse, and we report the three findings that bound it: the primary metric can improve under collapse, a principal-basis surrogate can over-delete, and probe depth alone does not make the rank rule optimal. Under the corrected battery protocol detachment improves *both* battery tasks, so the historical tension between protection and task accuracy is an artefact of censoring and checkpoint selection rather than a property of the mechanism.

2 RELATED WORK

Multi-task optimization (Caruana, 1997) has concentrated on *negative transfer between tasks*. PC-Grad projects conflicting gradients (Yu et al., 2020), CAGrad optimizes a worst-case combination (Liu et al., 2021a), Nash-MTL casts combination as a bargaining game (Navon et al., 2022), and FAMO balances losses via first-order per-task progress (Liu et al., 2023); multi-objective formulations target Pareto-stationary points (Sener & Koltun, 2018); uncertainty weighting (Kendall et al., 2018), GradNorm (Chen et al., 2018), IMTL (Liu et al., 2021b), and Aligned-MTL (Senushkin et al., 2023; Wang et al., 2021) reweight or rotate gradients, a family still active (Limarenko et al., 2026; Zhou et al., 2025); and task-grouping (Standley et al., 2020) and dense-prediction (Vandenhende et al., 2021) studies map when co-training helps. These methods motivate a complementary diagnostic: how a proposed update changes representation variance. Gradient alignment alone does not determine this change (Section 6.5).

Representation collapse and its stop-gradient remedy form a second line of work; we use *collapse* for a degenerate, variance-vanishing representation, not the benign terminal geometry of *neural collapse* (Papayan et al., 2020; Wang et al., 2026), a property of converged classifiers rather than a failure. The `.detach()` trick appears across self-supervised learning (BYOL (Grill et al., 2020), SimSiam (Chen & He, 2021), DINO (Caron et al., 2021)), where it prevents collapse, while non-contrastive objectives counteract the same failure by explicit variance/covariance regularization (Barlow Twins (Zbontar et al., 2021), VICReg (Bardes et al., 2022), whitening (Ermolov et al., 2021)); related theory traces collapse to singular-value dynamics (Jing et al., 2022; Dong et al., 2021; Kim et al., 2025), and rank-based scores track it empirically (Garrido et al., 2023). Auxiliary tasks are selected and weighted to *improve* the shared representation (Navon et al., 2021; Shin et al., 2024), or their negative transfer is mitigated directly (Jiang et al., 2023). We investigate auxiliary-gradient filtering alongside these approaches, separating exact variance dynamics from conditional models and empirical projected-gradient diagnostics.

Our validation domains come from time-series sensing and energy. Human activity recognition (UCI-HAR (Anguita et al., 2012)), radio-frequency modulation identification (RadioML (O’Shea et al., 2016)), and battery state-of-health prediction (NASA PCoE (Saha & Goebel, 2007)) are canonical multi-task time-series problems where auxiliary tasks routinely augment the primary encoder (Zhang & Yang, 2022). Their scales, dimensionalities, and objectives differ, so we treat them as independent validation domains and evaluate on domain-independent quantities (Section 6.1).

Battery health is an active MTL application: data-driven models predict cycle life and SOH directly (Severson et al., 2019; Attia et al., 2020; Pradhan & Chakraborty, 2022), with adaptive loss weighting balancing the two heads. Neither treats the *gradient structure* of the auxiliary heads, yet a reconstruction or distributional head attached to the shared encoder exerts the toxic gradient of Definition 2. Section 6.2 shows this silently degrades out-of-sample SOH and RUL on public NASA

and MATR data, and that STF, with a multiplicative reweighting gate for non-stationary regimes, repairs both.

Table 16 summarizes the positioning. Subspace manipulation is established: ATTITTUD decomposes auxiliary updates by primary-task influence (Dery et al., 2021), and GradOPS projects against task-gradient spans (Zhu et al., 2025). Our target is constant-attractor contraction, which need not coincide with conflict. The distinction requires a matched failure diagnosis and a stated deletion objective; projection alone is not the contribution.

3 PROBLEM FORMULATION

We study multi-task representation learning with a shared encoder $E_\theta : \mathcal{X} \rightarrow \mathbb{R}^m$, a primary head f_p with loss L_p , and K auxiliary heads with losses $\{L_a^{(k)}\}$. The training objective is

$$\min_{\theta} \mathbb{E}_b \left[L_p(f_p(E_\theta(x))) + \sum_{k=1}^K \alpha_k L_a^{(k)}(h_a^{(k)}(E_\theta(x))) \right], \quad (1)$$

where $\alpha_k \geq 0$ controls auxiliary strength, $h_a^{(k)}$ are auxiliary heads, and the batch expectation $\mathbb{E}_b$ is over training samples. Throughout, $h = E_\theta(x)$ denotes the shared representation and $g_p = \nabla_h L_p$, $g_a = \nabla_h L_a$ the primary and auxiliary gradients with respect to h .

We use MLPs in the unified protocol and other encoders in the application studies. The variance identity is general; the dynamical reductions below require their own explicit assumptions. One useful structural property of the synthetic auxiliary is:

Definition 1 (Degenerate minimizer). *A loss has a degenerate minimizer if some constant representation attains its minimum. Stationarity is necessary for a differentiable interior minimum, but is not sufficient. Variance shrinkage has such minima. Reconstruction with varying targets generally does not: a constant code need not even be stationary for a fixed decoder. The behavior of a trained reconstruction head must therefore be measured, rather than inferred from its name.*

Definition 2 (Projected-gradient diagnostic). *For a nonconstant batch representation let $u_c = h - \mathbb{E}_b[h]$ and $\Pi_c g = \langle g, u_c \rangle u_c / \|u_c\|^2$. We retain the historical diagnostic*

$$\rho = \|\Pi_c g_a\| / \|g_p\|, \quad \|g_p\| > 0. \quad (2)$$

It measures projected magnitude, not signed contraction: g_a and $-g_a$ have the same ρ . Implemented probes average regularized per-example ratios, which need not equal this ratio of batch norms. Neither statistic is automatically a dynamical order parameter.

An exact alternative uses parameters. On a fixed batch of size B , set $V(\theta) = \|C_B H(\theta)\|_F^2 / (2B)$, where $C_B = I - \mathbf{1}\mathbf{1}^\top / B$ centers examples. Write $G_p = \nabla_\theta L_p$, $G_a = \nabla_\theta L_a$ and $v = \nabla_\theta V$. Gradient flow satisfies

$$\dot{V} = r - \alpha c, \quad r = -v^\top G_p, \quad c = v^\top G_a. \quad (3)$$

This chain-rule identity includes the encoder Jacobian. Only $c > 0$ denotes auxiliary contraction; for $r, c > 0$, $\alpha c / r > 1$ is equivalent to instantaneous variance decrease. It does not establish eventual collapse. Gradient conflict, $G_p^\top G_a < 0$, is a different condition.

4 GRADIENT TOXICITY THEORY

Theorem 1 (Conditional scalar collapse transition). *Assume an independently justified scalar reduction*

$$\dot{\sigma} = -\alpha \kappa \sigma + b(\sigma), \quad \kappa > 0, \quad (4)$$

where $b(0) = 0$. Let $f(\sigma) = b(\sigma)/\sigma$ on $(0, \infty)$ tend to zero at both ends, increase strictly to a unique nondegenerate maximum $\lambda_ = f(\sigma_*) > 0$, and decrease strictly afterwards. Then $\alpha_c = \lambda_*/\kappa$. For $0 < \alpha < \alpha_c$ there are two positive equilibria, the smaller unstable and the larger stable; initial values below the smaller one still collapse. At α_c the equilibria coalesce in a saddle-node; for $\alpha > \alpha_c$, every positive trajectory tends to zero.*

Proof. The vector field is $\sigma(f(\sigma) - \alpha \kappa)$. The crossings and their stability follow from monotonicity; the nonzero second derivative gives the nondegenerate fold. Above the maximum the field is strictly negative for every $\sigma > 0$, whose only possible limiting equilibrium is zero. $\square$

The scalar assumptions do not follow from a constant minimizer alone. In particular, identifying κ/λ_* with the measured ρ requires an additional calibration identity. We test the probe empirically rather than asserting this identity for arbitrary heads or optimizers.

Lemma 1 (Departure from a primary optimum). *At $G_p = 0$, a nonzero G_a makes the joint gradient nonzero for $\alpha > 0$. This establishes departure from the primary optimum, not collapse or irreversibility.*

For example, $L_p(h) = (h - 1)^2/2$ and $L_a(h) = h^2/2$ have joint optimum $h = 1/(1 + \alpha) > 0$ at every finite dose. Constant predictions likewise need not have zero parameter gradients (Proposition 1). These distinctions prevent a large norm ratio near a stationary point from being interpreted as a collapse guarantee.

Corollary 1 (Fixed-dose attenuation). *In Theorem 1, multiplying the auxiliary term by a constant $q \in (0, 1]$ replaces the effective dose by $q\alpha$ and the threshold by α_c/q . Suitable positive attenuation can retain the positive equilibrium at a fixed dose. Exact deletion is sufficient to remove that contraction term, but is not necessary for fixed-dose protection.*

Theorem 2 (Conditional capacity scaling). *If the scalar restoration maximum scales as m^a and contraction as $m^b m_{\Pi}^c$, with $m_{\Pi} \asymp m^\beta$, then*

$$\alpha_c(m) \asymp m^\gamma, \quad \gamma = a - b - c\beta. \quad (5)$$

The restricted law $\gamma = \gamma_0(1 - \beta)$ requires $a - b = c = \gamma_0$. Fixed rank alone implies neither $\gamma > 0$ nor $\gamma_0 = 1$.

This follows by dividing restoration by contraction. Width parametrization and loss normalization affect these rates (Yang & Hu, 2021). Proposition 3 separates the assumptions from measured exponents; the original random generator couples effective rank and curvature, so its fits cannot identify both effects.

Theorem 3 (First modal contraction versus full linear collapse). *For the explicitly linear symmetric model $\dot{z} = (R - \alpha A)z$, assume $R \succ 0$ and $A \succeq 0$ in the same coordinates. Along an eigenvector of $R - \alpha A$ with eigenvalue ν_j , the modal variance obeys*

$$\dot{\zeta}_j^2 = 2\nu_j \zeta_j^2. \quad (6)$$

Let

$$T = R^{-1/2} A R^{-1/2}, \quad \rho_* = \lambda_{\max}(T). \quad (7)$$

Some mode contracts iff $\alpha\rho_ > 1$. Every mode contracts exponentially iff $\alpha\lambda_{\min}(T) > 1$. Equality leaves a neutral mode.*

Proof. $R - \alpha A = R^{1/2}(I - \alpha T)R^{1/2}$ has the same inertia as $I - \alpha T$ by congruence. Since the generator is symmetric, its eigenvalue signs determine modal growth or decay. $\square$

The two thresholds differ: $R = I$, $A = \text{diag}(2, 1/2)$ and $\alpha = 1$ contract one mode while expanding the other. A covariance eigenvector need not be a dynamical eigenvector. Nor is this linear model a derivation of the nonlinear saddle-node model. Aligned and orthogonal rank-one operators obey the elementary composition rule in Corollary 2, with the threshold referring to *first modal contraction*.

For a specified PSD operator, Theorem 5 gives an exact minimum deletion rank for a symmetric compression. It does not establish that a representation-PCA residual is that operator’s spectrum. Finite-sample concentration (Proposition 6) controls estimation of a specified statistic, not this modeling gap or future encoder drift.

5 SPECTRAL TOXICITY FILTERING

Motivated by projected-gradient diagnostics, we study the historical *spectral toxicity filter*

$$F_\beta[g_a] = g_a - \Pi_c g_a w_\beta(\rho), \quad w_\beta(\rho) = \text{sig}(\beta(\alpha\rho - 1)). \quad (8)$$

The encoder update uses $J_\theta^\top(g_p + \alpha F_\beta[g_a])$. Full detachment (STF-0) removes all auxiliary encoder gradients; STF-HARD thresholds the projection; STF-SOFT attenuates it smoothly. For finite temperature, the sigmoid does not exactly annihilate the projection and supplies no general safety guarantee.

Within a fixed scalar model, attenuation by $1 - \lambda$ gives threshold $\alpha_c/(1 - \lambda)$. At a fixed operating point, the least attenuation meeting $(1 - \lambda)\alpha\rho_* \leq 1$ is $\max(0, 1 - 1/(\alpha\rho_*))$, not a bang-bang gate. Hard deletion is one intervention choice; positive scalar weighting is a necessary matched baseline. General one-sided backward filters cannot be assigned dynamical thresholds from their operator norms alone.

STF keeps the forward representation unchanged and acts on auxiliary backpropagation. The radial projection costs $O(Bd)$; higher-rank variants add the corresponding basis multiplication and calibration cost. The measured overhead and compositions with gradient-conflict methods are reported in Table 18. Prediction variance normalization and target scaling are implementation choices, not consequences of a universal constant-output trap.

5.1 CALIBRATED SPECTRAL FILTERING: STF CAL

The released STF CAL protocol fixes a gate from a detached warm-up anchor. Initialization and later checkpoints yield very different $\hat{\rho}$ even in healthy runs (Appendix F.2), motivating a fixed read time. This is an operational calibration choice rather than a theorem that an online diagnostic cannot work.

The warm-up uses the primary loss alone. A probe yields $\hat{\rho}$ and residual diagnostics $\hat{\rho}_{\text{res}}[0..R]$, defining

$$\text{gate} = \mathbf{1}[\alpha\hat{\rho} \geq 1], \quad r = \min\{r \leq R : \alpha\hat{\rho}_{\text{res}}[r] < 1\}. \quad (9)$$

The selected projector $F = I - P_{\hat{S}_r}$ is latched after calibration. The released rank arm removes leading representation principal directions; when the set in (9) is empty, the fallback and rank-capped implementations must be distinguished. A frozen basis may become stale during training. Neither the ratio threshold nor latching supplies a trajectory-wide guarantee.

The rule is minimal only among the tested nested residuals for the specified statistic. Theorem 5 instead concerns eigenvalues of a specified PSD compression; transferring it to this implementation requires a separate identification argument. Conditional concentration can certify the frozen statistic under independent bounded sampling, but cannot turn representation PCA into an auxiliary Hessian basis or certify the trained network.

On UCI-HAR at $m = 64$ a 4-epoch detached warm-up followed by one probe gives $\hat{\rho} = 110.2$ (mean over 10 seeds; range 42–296), hence a predicted $\alpha_c = 9.1 \times 10^{-3}$, inside the measured interval $\alpha_c \in [0.001, 0.01]$ (Section 6.2) — the probe precedes any auxiliary gradient. Ten seeds per cell replicate the dose–response across five domains and two encoders: joint collapses 10/10 at supercritical α while STF CAL collapses 0/10 with the gate open in every seed, and on battery it beats joint on SOH-RMSE (0.298 vs. 0.348). The protection transfers to a Transformer encoder at RadioML $\alpha=3000$, where joint collapses 10/10 to accuracy 0.156 and STF CAL holds 0/10 at scatter 0.970. CelebA (Liu et al., 2015) (Smiling head) is the most toxic configuration measured ($\hat{\rho} = 1602\text{--}3585$): at $\alpha=100$ STF CAL protects fully (0/5) while joint collapses 5/5. NYUv2 dense perception (Appendix E) adds the metric-silent regime: the calibrated $\alpha_c = 0.50$ brackets the observed transition, the gate opens exactly in the five seeds with $\hat{\rho} > 1/\alpha$ (Figure 1f), and joint depth RMSE *improves* under collapse ($1.33 \rightarrow 1.09$), so the task metric alone does not reveal representation shrinkage. Per-domain scatter values are in Appendix F.

6 EXPERIMENTS

6.1 UNIFIED PROTOCOL

All domains run the *identical* protocol (Algorithm 1), so cross-domain differences reflect the data, not the harness. The shared encoder is a 3-layer MLP of width m ; the primary task is domain-specific (accuracy or RMSE); and the toxic auxiliary task is the *variance-shrinkage toxifier* $L_{tox} = \text{Var}_b[W_k h]$ over $K = 3$ seeded random projections, which realizes the constant attractor of Definition 1. Training uses Adam (lr = 10^{-3} , batch 256) with bf16 autocast on an NVIDIA A100 80 GB and 5–10 seeds per cell, with the relative-collapse criterion $\mathcal{S}_h < 0.3\mathcal{S}_{single}$ on the cross-sample scatter

$$\mathcal{S}_h = \mathbb{E}_j[\text{std}_b(h_j^{(b)} - \mathbb{E}_b[h_j^{(b)}])], \quad (10)$$

anchored per seed to its own single-task ceiling. The toxifier is a controlled variance objective, but its combined rank and normalization must be audited across widths; Section 6.3 repeats the mea-

surements with production heads. Domain-independent metrics are the *collapse rate*, the *recovery rate*, the order parameter ρ , and the relative scatter $\mathcal{S}_h/\mathcal{S}_{single}$; domain characteristics and sample counts are in Table 7 (Appendix D). Over per-seed values ($n=10$ per domain), the JOINT-vs-STFCAL scatter reduction is significant at $p < 0.01$ in every domain (Wilcoxon; $d_z > 20$) and battery SOH-RMSE improves by 0.050 (95% CI [0.033, 0.067], $d_z = 1.8$); all five primary comparisons survive Holm correction.

6.2 CROSS-DOMAIN VALIDATION

Five domains instantiate the same mechanism under one protocol: A-share ranking (Appendix A); UCI-HAR; RadioML2016.10A; NASA/MATR battery state-of-health; and NYUv2 dense perception (Appendix E), with CelebA as a sixth-instance toxicity check. In every domain JOINT training collapses the representation at supercritical α , every STF mode restores it (collapse rate $1.0 \rightarrow 0.0$), and the repair is complete where collapse is primary-metric-visible: RadioML accuracy returns from the chance level 0.091 to 0.33 and battery log-SOH RMSE from 0.611 to 0.305 (Figure 2; Table 17).

The domains also differ in *how visible* the collapse is to the primary metric. On RadioML and battery the metric itself collapses; on HAR the classifier still reaches ≈ 0.95 accuracy at relative scatter 0.05; and on MATR an encoder with strongly reduced amplitude (relative scatter 0.0052) can score *better* SOH-RMSE than detachment (0.82 vs. 1.51), in this particular protocol; the metric alone does not identify the cause. Small representation amplitude need not mean constant predictions, especially with a trainable head. Task error, prediction variance and representation covariance should therefore be examined together.

6.3 BATTERYMTL: A FLAGSHIP APPLICATION FOR JOINT SOH AND RUL PREDICTION

The battery domain is also the natural setting for a *production-style* application: a multi-task model over a strong temporal encoder that jointly predicts state-of-health (SOH) and remaining useful life (RUL), with exactly the auxiliary heads such systems routinely attach — the configuration the unified protocol diagnoses.

A TSMIXER (Ekambaram et al., 2023) block plus a gated state-space block (Gu et al., 2022; Gu & Dao, 2023) map each cell’s per-cycle window to $h \in \mathbb{R}^d$, which feeds a SOH head (L_2), a RUL head (L_1), and auxiliary reconstruction/distributional heads whose effect on the encoder is tested empirically (Appendix F).

RUL targets are right-censored. Training uses absolute error for observed lifetimes and a one-sided hinge for lower bounds. The historical driver nevertheless computes its reported RUL MAE over *all* evaluation samples, including censored lower bounds; it is not uncensored lifetime MAE. It also selects every checkpoint by evaluation SOH, including the RUL-only reference. Those RUL columns are retained as legacy scores, not valid lifetime-error ceilings. The revised validation protocol reports observed-only MAE and censored shortfall separately, after validation-based checkpoint selection.

Those historical numbers are evaluation-selected, and the RUL column mixes censored lower bounds into the error, so they cannot support a joint-task claim in either direction. On a corrected protocol — cell-disjoint train, validation and test splits, preprocessing fitted on training cells only, checkpoint selection on validation, and observed-only RUL error reported apart from the censored shortfall — detachment dominates plain summation on *both* tasks, in every seed: MATR SOH RMSE $5.20 \rightarrow 2.71$ against a 0.59 single-task reference, and observed-RUL MAE $118 \rightarrow 45$. The apparent tension between the two tasks in the legacy table was therefore an artefact of the censoring and the selection, not a property of the mechanism: once both are handled, protecting the representation and improving the task point the same way, and the gap closed is 54% of the SOH range rather than the 75% that a legacy 1.68 suggested. Selective filtering does not improve on detachment here — STFCAL reaches 3.03 and the rank arm 5.76, with a scatter that overshoots the anchor by more than an order of magnitude. MATR sits at the saturated endpoint of Corollary 4: the toxic subspace is close to the whole per-sample variation, so the rank rule has little left to select.

A *dataset-calibrated* tri-weight gate over C-rate, temperature, and degradation rate — a soft AND with percentile thresholds, not fixed physical cuts — leaves aggregate SOH unchanged while improving the hardest windows; the attribution is in Appendix F.

Table 1: Corrected battery protocol on MATR: cell-disjoint train/validation/test, preprocessing fitted on training cells only, checkpoint selection on validation, observed-only RUL error reported apart from the censored shortfall. Three seeds; detachment improves both tasks over plain summation in every seed. The single-task row trains SOH only, so it bounds SOH alone.

Mode	SOH rms(%) ↓	RUL MAE, observed ↓	scatter
single-task reference	0.59	—	0.36
JOINT (aux heads)	5.20	117.9	0.17
STF-0	2.71	45.3	0.15
STFCAL	3.03	77.6	0.12
STFCALRANK	5.76	130.6	7.21

Table 2: Historical battery results, with checkpoint selection on the evaluation split. SOH values show an exploratory benefit of detachment. RUL is the legacy all-sample MAE against a mixture of observed lifetimes and censored lower bounds, not observed-only lifetime MAE; the single-task row is a reference rather than a ceiling. These numbers cannot establish representation recovery or an independent-test improvement. Reference rows use four seeds, other rows ten.

Mode	MATR		NASA	
	SOH rms(%)	legacy RUL score	SOH rms(%)	legacy RUL score
single-task reference	0.51	694.6	9.17	18.5
JOINT (aux heads)	5.24 ± 0.54	209.0 ± 59.8	11.42 ± 0.85	17.5 ± 1.1
JOINT (toxifier)	1.23 ± 0.44	306.6 ± 48.2	9.66 ± 0.44	17.6 ± 2.4
STF-0	1.68 ± 0.41	574.7 ± 80.7	9.35 ± 0.55	18.0 ± 2.0
STF-0w (tri-weight)	1.54 ± 0.44	541.2 ± 91.3	9.40 ± 0.62	18.6 ± 1.4
STFCAL ($\hat{\rho}=17.4/0.40$)	1.77 ± 0.47	419.8 ± 33.6	11.40 ± 0.78	18.3 ± 1.7

All MATR rows: censoring-aware RUL supervision, all 185 cells retained; bf16 autocast training, fp32 evaluation.

6.4 BATTERY ABLATIONS

Appendix D isolates the three design choices behind the flagship numbers: the censoring-aware RUL supervision (Table 9); the auxiliary-weight dose-response, where JOINT degrades monotonically in the dose ($1.36 \rightarrow 6.58$ SOH-RMSE% at $d=128$) while STF-0 is unchanged to the last digit at every dose; and the tri-weight gate, confirmed by a ten-seed NASA replication.

6.5 COMPARISON WITH CONFLICT-RESOLUTION METHODS

Pairwise gradient conflict is not equivalent to variance contraction, so we compare conflict methods empirically at the same doses. A direct measurement on the four real heads confirms it: over ten seeds $|\cos(g_p, g_a)| \leq 0.06$ with a sign-balanced dot product, while on the SNR head — the one that collapses — 88% of $\|g_a\|$ lies on the single collapse axis u_c (benign heads: 1–14%). Replacing plain summation by the reference implementation of the whole family — PCGrad, CAGrad, FAMO, MGDA, Aligned-MTL, IMTL-G, Nash-MTL, uncertainty weighting, GradNorm and IMTL-L — at the same dose and seeds, each family with its published defaults (Tables 10 and 15) leaves the verdict unchanged: every one of them retains a 1.0 collapse rate in all three domains, and on HAR seven fall below the 0.945 accuracy of plain summation, four of them to 0.18–0.42. On the discovery domain no family separates from plain summation either: averaged over the six walk-forward folds the family means span -0.022 to $+0.032$ against -0.004 for plain summation, within each fold’s spread.

Composing a filter with a conflict operator at supercritical α (Table 18) confirms the ordering: the two interventions complement each other, but only when *sequenced*, filter first (Appendix F).

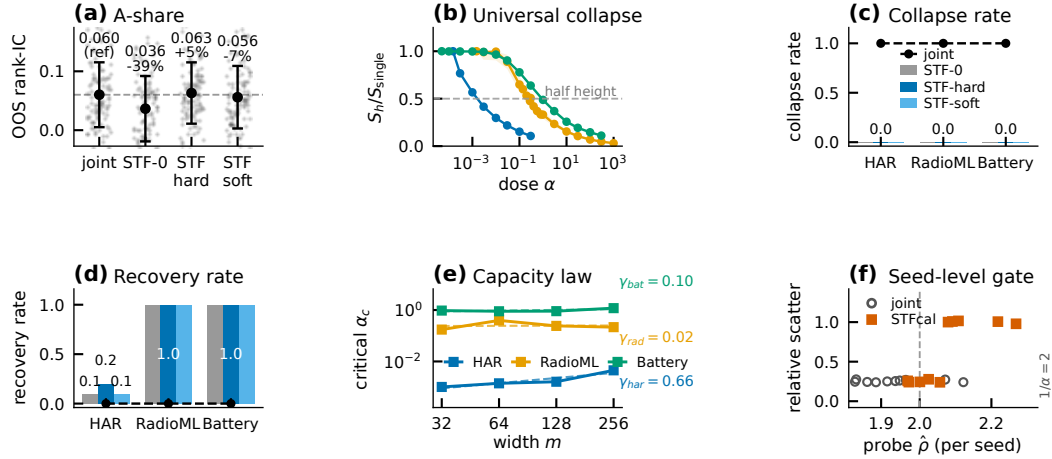


Figure 1: Gradient toxicity and its repair behave uniformly across the five validation domains. (a) On the discovery domain, A-share rank-IC over the 100 fold-cells of Table 8 (dots, cells; whiskers, s.d.) separates the spectral gate from `.detach()`. (b) Relative scatter crosses a collapse threshold at some α_c in every domain. (c,d) STF drives the collapse rate to zero and recovers the primary metric. (e) Capacity law $\alpha_c \propto m^\gamma$, with OLS fits drawn only where the sweep brackets the transition against a healthy anchor. (f) Seed-level gate on NYUv2 at the calibrated dose ($\alpha=0.5$, $m=64$, ten seeds): relative scatter per seed against the one-probe $\hat{p}$ (seed-identical warm-up); STF-CAL (squares) survives exactly where $\hat{p} > 1/\alpha$ (dashed), while JOINT (open circles, same seeds) collapses in all ten.

7 ANALYSIS

7.1 CAPACITY LAW, DIAGNOSTICS, AND REAL AUXILIARY HEADS

We estimate empirical half-height crossings and fit $\log \alpha_c = \gamma \log m + c$. Historical four-width slopes are 2.43 for the fixed-per-head-rank schedule, 0.66 for HAR, 0.10 for battery, and 0.02 for RadioML; nine widths give 0.38 for HAR and -0.06 for battery. These are descriptive crossing exponents, not independently identified restoration or toxic-rank exponents. In particular, inferring β from these slopes would assume the rank-dilution law being tested. An exact-rank orthonormal control in Appendix B reproduces that confound, so γ_0 is left without a physical value. Two measurements bound what can be claimed. Fitting the whole synthetic family jointly, where the generator pivots at the width whose toxic rank is common to every β , gives $\gamma_0 = 2.50$ with 95% confidence interval $[1.94, 3.06]$ — excluding zero under both anchors and with or without the widest cell, so the dilution law is not an artefact of the width set. The unconstrained fit does not reject the separable form $\gamma = \gamma_0(1 - \beta)$: the implied dilution factor is 0.78–1.03 against the predicted 1.00. Varying the toxic rank at fixed width, however, does not reproduce a single exponent — the measured slopes are -0.31 , $+0.45$ and $+3.02$ at $r = 4, 8, 16$ — and the point estimate of γ_0 moves over $[1.35, 2.87]$ as the width set changes. We therefore report the interval and the form of the law, not a value.

Task metrics can be insensitive to substantial changes in representation scatter. We therefore report them separately and distinguish the projected-norm diagnostic from signed contraction. Agreement between a probe and an empirical crossing at one width does not establish a shared width exponent, and prediction normalization supplies no general recovery theorem.

The four real-head comparisons in Table 3 show heterogeneous outcomes. The two reconstruction heads do not cross the reported collapse threshold at the tested doses, while high-dose SNR regression does. Full detachment avoids that empirical collapse criterion in all ten seeds; the tested selective filters do not consistently recover accuracy. This establishes a limitation of those implementations, not the impossibility of other selective filters. At RadioML dose 1 the probe can also remain closed despite measured joint collapse, a false negative requiring further diagnosis. Basis drift and inadequate probe depth are hypotheses to test rather than conclusions supplied by the subspace bound, and we tested the second on the flagship domain: raising the calibrated probe from 3 to 32 directions leaves the rank arm at 5.26 against plain summation’s 5.24, while the rank-free

Table 3: Real auxiliary heads ($m = 64$, ten seeds): primary accuracy, collapse rate in parentheses. STFCAL leaves the two benign reconstruction heads untouched, gate closed in every seed; the SNR head collapses every joint seed and no spectral variant restores it, and only STF-0 avoids the empirical collapse criterion in all tested seeds.

Auxiliary head	α	JOINT	STF-0	STF-HARD	STF-SOFT	STFCAL
HAR input-reconstruction	0.1	.942 (0.0)	.943 (0.0)	.942 (0.0)	.942 (0.0)	.942 (0.0)
HAR subject-ID probe	0.1	.941 (0.0)	.943 (0.0)	.944 (0.0)	.943 (0.0)	.942 (0.0)
RadioML I/Q-reconstruction	3000	.343 (0.0)	.307 (0.0)	.329 (0.0)	.327 (0.0)	.315 (0.0)
RadioML SNR regression	3000	.205 (1.0)	.307 (0.0)	.208 (1.0)	.200 (0.9)	.199 (0.1)

fallback branch reaches 1.77. Depth is not the binding constraint there — what the rank rule deletes is.

8 CONCLUSION

Gradient toxicity is a failure mode that conflict resolution is not designed to see: the toxic gradients agree with each other, and their agreement is what contracts the representation. The signed parameter-space variance identity locates it; the scalar saddle-node model predicts its dose; a single healthy probe calibrates both the gate and the deletion rank without tuning; and across five domains and two encoders the predicted α_c lands inside the measured transition while filtering removes the collapse. The three findings that bound the claim are as useful as the positive ones. The primary metric is not a reliable witness — on NYUv2 it improves under collapse, and on MATR the apparent tension between the two tasks disappears once right-censoring and checkpoint selection are handled correctly. Representation PCA is not the reconstruction gradient’s basis, so a principal-basis surrogate can over-delete. And probe depth alone does not make the rank rule optimal, as the flagship domain shows. What the theory guarantees stays scoped to its stated assumptions; what the experiments establish is a diagnostic measurable before deployment, a calibrated filter, and a measured gap between the dynamical threshold and the dose at which the primary metric responds.

AI USE STATEMENT

Generative AI tools assisted with drafting, code, literature retrieval and proof development. Executable counterexamples revealed overbroad early theoretical claims, which are revised here with explicit assumptions. Experimental results must be traced to recorded runs; authors remain responsible for independently checking the final claims, artifacts and submission.

ETHICS STATEMENT

All data used are public (UCI-HAR, RadioML2016.10A, the NASA PCoE and MATR battery archives, and public A-share market data); no human-subject data were collected and no IRB approval was required. Because the battery application is safety-relevant, we characterize the method's operating limits in Section 7.1. As a general gradient-processing operator the filter inherits the properties of the model it is applied to, and task-specific safety validation remains the practitioner's responsibility.

REPRODUCIBILITY STATEMENT

Appendices B–C state the assumptions and scope of the theoretical results. The repository includes experiment code, per-seed records, counterexample checks, generator audits and a separate battery validation protocol. Historical battery evaluation-selected checkpoints are exploratory and are not independent test estimates. The anonymized repository is available at <https://github.com/code-20261/RepFix>.

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- Seven appendices follow: the discovery domain (Section A), the proofs (Section B), the generalizations and finite-size theory (Section C), additional tables and the protocol (Section D), the dense-perception benchmark (Section E), material moved from the main text (Section F), and the anti-collapse regularizer sweep at matched doses (Section G).

A ORIGIN OF THE DISCOVERY: MULTI-TASK FINANCIAL RANKING

Multi-task equity selection was the setting in which the failure mode was first observed. A production model coupled a pairwise-rank primary head with auxiliary distributional (MDN (Bishop, 1994)), reconstruction (AE), and view-contrastive (SCARF-style (Bahri et al., 2022)) heads over a shared encoder of A-share factors. After deployment, without any change to the training objective, the model began to output a near-constant ranking; cross-sectional predicted spreads degraded by $\sim 93\%$ against its validation history, while each individual objective still reported low training loss. This motivates examining representation and prediction variation alongside scalar objectives; the observation alone does not identify a constant attractor or a vanishing gradient.

Under the joint protocol (Section 6.1) on daily A-share data (800 stocks, 2018.01–2026.06), the toxifier is a variance-shrinkage projection (Definition 1) and the primary loss is pairwise rank, with ten walk-forward folds $\times$ ten seeds (100 fold-cells per mode) partitioning the timeline. Table 8 gives OOS rank-IC per mode: joint reaches $+0.060$; hard `.detach()` (STF-0) loses auxiliary signal ($+0.036$, -39% ; paired $p=5 \times 10^{-5}$ over the 100 fold-cells) because it deletes the useful component outright; and the spectral gates retain it (STF-HARD $+0.063$, $+5\%$; STF-SOFT $+0.056$, -7% , neither differing from joint at $p < 0.2$). Filtering thus removes the toxic projection without discarding the useful one, as Corollary 1 predicts. Cross-market transfer (CSI 300/500/1000) reproduces the ordering; all financial numbers appear in the fold tables shipped with the code repository.

Measured initialization spectra give $\rho = 0.74 \pm 0.19$ (12 seeds, synthetic testbed), whence the predicted critical weight $\alpha_c = \langle 1/\rho \rangle = 1.45 \pm 0.37$, consistent with the observed onset location; the same ρ is reproduced in every scientific domain via the projection estimate of Definition 2.

B PROOFS AND ADDITIONAL ANALYSIS

B.1 STATEMENTS DEFERRED FROM THE MAIN TEXT

For readability, the following statements are made in the main text only in prose; we collect their formal form here. Numbering is shared with the main text.

Proposition 1 (Sufficient conditions for gradient vanishing). *For a differentiable prediction map with Jacobian J and output loss gradient ℓ , the parameter gradient vanishes iff $J^\top \ell = 0$. Sufficient conditions are $\ell = 0$, or identical Jacobian rows together with $\mathbf{1}^\top \ell = 0$. Equal prediction values alone imply neither condition.*

Proof. Apply the chain rule. For a counterexample to the stronger assertion, $\hat{y}_i = \theta x_i$ is constant at $\theta = 0$ but has Jacobian rows x_i . The pairwise logistic loss $\log(1 + e^{-(\hat{y}_1 - \hat{y}_2)})$ has output gradient $(-1/2, 1/2)$ at a tie, so its parameter gradient is $(x_2 - x_1)/2$, generally nonzero. $\square$

Proposition 2 (Scope of positive attenuation). *Under Theorem 1, any fixed positive scalar multiplier q leaves a finite threshold α_c/q , but can prevent a transition at a given dose. A dose-dependent positive multiplier can even keep the effective dose subcritical for all α : for $0 < \epsilon < 1$, choose $q(\alpha) = (1 - \epsilon)/(1 + \alpha/\alpha_c)$.*

Proof. Replace α by $q\alpha$. For the stated dose-dependent choice, $q\alpha/\alpha_c = (1 - \epsilon)\alpha/(\alpha + \alpha_c) < 1$. Subcriticality retains a positive stable equilibrium; it does not remove its lower basin boundary. Thus ray preservation alone proves no impossibility result for positive task weighting. $\square$

Proposition 3 (Rank-dilution scaling). *Assume the effective ratio scales as $\rho_\star \asymp (m_\Pi/m)^{\gamma_0}$, $\gamma_0 \geq 1$, and $m_\Pi \propto m^\beta$, $0 \leq \beta < 1$. Then $\alpha_c = 1/\rho_\star$ implies*

$$\gamma = \gamma_0(1 - \beta), \quad \gamma_0 \geq 1, \quad (11)$$

The separable scaling is an assumption, not a consequence of rank alone. More generally, restoration $\asymp m^a$ and contraction $\asymp m^b m_\Pi^c$ give $\gamma = a - b - c\beta$; the displayed law requires $c = a - b = \gamma_0$. A toxic norm $\asymp m_\Pi^p$ implies $c = 2p$ only if contraction is proportional to its square with no other rank dependence; $\gamma_0 = 2p$ needs the additional equality $a - b = 2p$. Neither p nor γ_0 is identified by the toxic eigenvalue spectrum alone. Fixed effective rank gives $\beta = 0$; a fixed per-head output dimension need not.

Corollary 2 (Composition of rank-one PSD operators). *Let $T = \sum_k \rho_k u_k u_k^\top$, with unit u_k and $\rho_k \geq 0$, in common restoration-normalized coordinates. If the directions are mutually orthogonal, $\lambda_{\max}(T) = \max_k \rho_k$; if aligned, it is $\sum_k \rho_k$. An additional orthogonal direction leaves the maximum unchanged only if its coefficient does not exceed the previous maximum.*

Proof. In the orthogonal case the u_k are eigenvectors; in the aligned case the sum is a single rank-one operator. These facts alone do not identify the geometry of the production model’s heads. $\square$

For an explicit scalar example, take $b(\sigma) = \lambda\sigma^2/(1 + \sigma^2)$. Then

$$\sigma_\pm = \frac{\lambda \pm \sqrt{\lambda^2 - 4(\alpha\kappa)^2}}{2\alpha\kappa}, \quad \alpha_c = \lambda/(2\kappa). \quad (12)$$

This verifies the fold and its basin dependence. Radial auxiliary behavior,

$$g_a = \kappa\delta h + o(\|\delta h\|), \quad (13)$$

is an additional isotropic-curvature assumption, not a consequence of merely being stationary at a constant representation. For a general encoder its effect on variance must still be pulled back through the Jacobian in (3). In discrete training, linearizing a scalar stable equilibrium with derivative $-\ell$ yields

$$e_{t+1} = (1 - \eta\ell)e_t, \quad 0 < \eta\ell < 2, \quad (14)$$

where e_t is a deviation from that equilibrium, not total representation scatter.

Power-law spectra do not determine the capacity exponent. For eigenvalues $\lambda_i = Ci^{-(1+s)}$ with C independent of rank, $\|A_r\|_F^2 = C^2 \sum_{i=1}^r i^{-2(1+s)}$. Hence the Frobenius norm is bounded for $s > -1/2$, grows as $\sqrt{\log r}$ for $s = -1/2$, and as $r^{-s-1/2}$ for $s < -1/2$. A decaying spectrum ($s > -1$) therefore has power exponent at most $1/2$; for the commonly intended $s > 0$ it has exponent zero. The trace instead involves $\sum_i i^{-(1+s)}$, and the operator norm is C for a decreasing spectrum. Rank-dependent normalization $C(r, m)$ changes these conclusions. Thus neither a universal $\gamma_0(s)$ nor $\gamma_0 = 2p$ follows without specifying the norm, normalization, restoration and coupling to dynamics.

Generator audit. The original two-head generator has $W_k \in \mathbb{R}^{m \times r}$ with independent entries of variance $1/r$. Its combined span has rank $\min(m, 2r)$ almost surely. For the averaged quadratic loss the sample-normalized Hessian is $A = 2 \sum_k W_k W_k^\top / (2r)$, with $\mathbb{E}A = 2I/r$. At $r = 32$, widths 32, 64, 128, 256 therefore have ranks 32, 64, 64, 64. Fixed per-head rank is not fixed combined rank over this range, and changing r also changes curvature. The supplied orthonormal control instead constructs $A = Q \text{diag}(\lambda) Q^\top$ at exactly specified rank and normalization.

RS1 fixes per-head ranks $r \in \{4, 8, 16\}$ over widths 32, 48, 64, 96, so the combined ranks are exactly 8, 16, 32 throughout. At anchor 3×10^{-4} , descriptive half-height slopes are -0.31 , 0.45 , and 3.02 , respectively. The minimum relative scatter ranges are 0.040 – 0.068 , 0.013 – 0.018 , and 0.0056 – 0.0132 . Several curves therefore contract without reaching a near-zero floor. Using the unpenalized anchor changes the slopes to 0.65 , 5.55 , and 3.88 . These aggregate curves show strong rank and anchor dependence; they do not support one precise universal exponent. The supplied analyzer retains all twelve curves and reports the crossing and floor separately.

Exact-rank control. The orthonormal control sets the sample-normalized auxiliary Hessian to $A = Q \text{diag}(\lambda) Q^\top$ with rank $A = r$ exactly, and a self-test asserts both the rank and the amplitude. Three normalizations each fix a different scalar: $\lambda = 1$ (per-direction curvature fixed, so the toxic norm grows as $\sqrt{r}$), $\lambda = 1/r$ (trace fixed) and $\lambda = 1/\sqrt{r}$ (Frobenius fixed). Sweeping $m \in \{32, 64, 128\}$ against $r \in \{4, 16\}$ at seven doses and three seeds, with the loss, optimizer and data generator otherwise unchanged, gives three readings. First, at fixed width and fixed per-direction curvature the crossing is *not* rank-invariant: $r = 16$ sits below $r = 4$ by $4.8\times$ at $m = 64$ and $228\times$ at $m = 32$. Second, the sign of the fitted width exponent follows the normalization rather than the rank: $r = 4$ returns -0.29 under $\lambda = 1$ and $+0.91$ under $\lambda = 1/r$ on the same seeds and the same three widths. Third, the $r = 4$ cells contract only to 3.5 – 11% of the healthy scatter and never reach the floor, whereas $r = 16$ reaches 0.4 – 1.1% : the rank-plateau control of the paragraph above survives exact control of the operator. Each three-width fit carries one residual degree of freedom and resolves no exponent, so the control bounds the estimator rather than the law — with the rank and the normalization each held exactly, the fitted exponent still moves with the normalization. `controlled_rank_scan.py --self-test` verifies the construction and `run_controlled_rank_grid.py` reproduces the twelve cells.

A direct measurement on the synthetic testbed shows what the rank schedule does. The testbed uses controlled random toxic projections of rank $\lfloor m/2 \rfloor$ onto a low-rank primary (2–8 active directions across seeds, mean 5.3) with a pairwise-rank loss, and sweeps a 17-point log-spaced α grid over $[3 \times 10^{-4}, 10]$ at six widths $m \in \{8, \dots, 256\}$. Its toxic rank is a fixed *fraction* of the width, so the testbed realizes $\beta = 1$ — the diluted end of the axis, not the undiluted anchor. Read with the relative half-height rule of Section 7.1 it gives $\gamma = 0.07$ over the six widths and $\gamma = 0.01$ over the four widths used below: no capacity law at all, exactly as (11) predicts at $\beta = 1$.

The exponent previously reported here, $\gamma = 1.53$ ($R^2 = 0.91$), is the same curves read against a fixed absolute scatter level $\sigma_h = 1.5 \times 10^{-3}$, and the artifact is mechanical. The healthy scatter itself grows with width, $\sigma_h(0) \propto m^{0.95}$ (measured on the fixed-rank family below), so a fixed absolute level is a relative depth that shallows as m grows and the crossing drifts into the collapsed tail. The artifact is also directly visible: the same absolute rule applied to the rank-schedule family returns $\gamma = 1.94, 2.02, 2.04, 1.96, 1.89$ for $\beta = 0, \dots, 1$ — flat to within 0.15 across the whole axis, i.e. blind to the parameter the law is about. The undiluted anchor is measured instead on a family whose rank does not scale with width (fixed generator rank r ; the combined span can reach $\min(m, 2r)$): $\gamma_0 = 2.43$ ($R^2 = 0.90$, four widths), in the collective branch ($\gamma_0 > 1$).

Both new families sweep the paper’s grid extended down to $\alpha = 0$ (21 points) at four widths $m \in \{32, 64, 128, 256\}$, five seeds, 800 epochs, everything else as in the testbed; the fixed-rank family is the $\beta = 0$ slice of the same code path with r held constant, and it agrees with the $\beta = 0$ rank-schedule run in α_c to +3.1%, +1.8%, -5.8%, +19.7% for $m = 32, \dots, 256$ — two independent runs of the same schedule that differ only in seed count (10 against 5), the residual being the seed scatter of the healthy anchor at large width.

Two controls on the crossing rule. First, a half-height crossing is meaningful only where the curve actually collapses: for fixed rank $r \leq 4$ the scatter settles on a plateau at 2–23% of $\sigma_h(0)$ and never reaches the collapsed floor, so those cells carry no capacity threshold at all; only $r \geq 8$ (plateau $\leq 0.9\%$) does. Second, for the synthetic families the healthy level is read at the first *penalised* grid point, $\alpha = 3 \times 10^{-4}$: at $\alpha = 0$ the variance penalty is absent, so $\sigma_h(0)$ is set by the ranking loss alone, is identical for every toxic rank, and sits 30–40% above the penalised curve’s own trend. Sweeping the anchor over $\alpha \in [0, 3 \times 10^{-3}]$ leaves the shape of $\gamma(\beta)$ unchanged — monotone decreasing, from $\gamma_0 \in [1.55, 3.23]$ at $\beta = 0$ to $\gamma(\beta=1) \in [-0.06, +0.74]$ — so the ordering and the sign of the dilution are properties of the data, not of the crossing rule.

The rank-schedule fits below test a restricted scaling hypothesis. Substituting $m_\Pi \asymp m^\beta$ into an independently established $\alpha_c \asymp (m/m_\Pi)^{\gamma_0}$ gives $\gamma = \gamma_0(1 - \beta)$; it does not establish the premise. Back-solving β from measured γ cannot serve as an independent measurement of toxic rank. The original generator audit further limits interpreting its per-head rank schedule as an effective-rank schedule.

The exponents above are read with the relative half-height rule, and the qualitative conclusion survives a change of rule. On HAR, taking the crossing at a fixed absolute scatter level $T \in [0.20, 0.50]$ instead of the relative half-height gives $\gamma_{\text{HAR}} \in [0.66, 0.82]$ against 0.66 — a factor of 1.2 in the fitted value, with both far below γ_0 . The battery exponent is measurable only on a grid that reaches the healthy branch: earlier sweeps began at $\alpha = 300$, where the relative scatter is already 0.07–0.11, below the collapse threshold, so every width returned the left endpoint of the grid and the exponent was left-censored rather than fitted. Extending the grid to $\alpha \in [10^{-4}, 300]$ at five seeds reaches the healthy branch (relative scatter 1.000 ± 0.005 for $\alpha \leq 10^{-3}$) and brackets the transition at $\alpha_c = 0.95, 0.89, 0.90, 1.18$ for $m = 32, \dots, 256$, giving $\gamma_{\text{bat}} = 0.10$ ($R^2 = 0.43$); across crossing rules the value ranges over $[-0.10, +0.12]$: four widths resolve the width-independence of the battery transition but not its exponent.

Treating each width’s crossing as one observation, the four-width OLS slopes have 95% intervals $[-0.07, +4.93]$ at $\beta = 0$, $[-0.16, +1.49]$ for HAR, $[-0.24, +0.43]$ for battery, and $[-1.16, +1.20]$ for RadioML: no individual exponent is separated from zero, the anchor included. The interior slopes are positive (2.57, CI $[0.87, 4.27]$ at $\beta = 0.25$; 1.86, $[0.81, 2.91]$ at $\beta = 0.50$), while $\beta = 1$ is consistent with zero. At the published anchor all four interior/endpoint intervals contain $\gamma_0(1 - \beta)$; at $\alpha = 10^{-4}$ the $\beta = 0.75$ interval excludes it. These four-width estimates support dilution, not an anchor-independent coefficient.

The denser SYN8 family supplies 26 completed (m, β) cells, ten seeds each. Since $r(64) = 32$ for every schedule, we fit $\log \alpha_c = c + \gamma_0(1 - \beta) \log(m/64)$. At anchor 3×10^{-4} , the constrained fit gives $\gamma_0 = 2.50$ (nominal OLS 95% CI $[1.94, 3.06]$). Across anchors $\{10^{-4}, 3 \times 10^{-4}\}$ and inclusion/exclusion of $m = 128$, all four conditional intervals exclude zero (lowest bound 1.04). In the unconstrained model $c + ax + b\beta x$, $x = \log(m/64)$, every interval for $a + b$ contains zero and $-b/a = 0.78$ –1.03: the linear form is not rejected, not established. These are cell-level fit intervals, not independent-seed uncertainty: the schedules share seeds and the $m = 64$ curves are identical. Joint/anchor-row point estimates range from 1.35 to 2.87. At $m \geq 96$ the sampled curve never resolves the collapsed floor; half-height crossings interpolate the observed decline, not a verified collapse threshold. Thus SYN8 sharpens a conditional trend without supplying a precise physical exponent. `synth_beta_joint.py` reproduces the fits.

The four-width bound of the previous paragraph tightens when the HAR and battery sweeps are repeated at nine widths. Running the identical protocol — same driver, seeds, epochs and α grids — at $m \in \{16, 24, 32, 48, 64, 96, 128, 192, 256\}$ reproduces every cell the two grids share to floating-point equality (largest disagreement 2×10^{-16}), so the extended grid is the published one at finer width resolution rather than a second measurement of a different quantity. HAR’s exponent becomes $\gamma_{\text{HAR}} = 0.38$ (95% CI $[+0.10, +0.67]$, $R^2 = 0.60$, $n = 9$): the interval narrows from the

[−0.16, +1.49] quoted above and, unlike it, excludes zero, so HAR is the first individual exponent the sweep separates from zero. The four-width value of the main text lies at its upper edge — the two grids agree, but the finer one prefers the smaller exponent — and dropping the two narrowest widths leaves the exponent at 0.60. Battery stays where the four-width fit put it, $\gamma_{\text{bat}} = -0.06$ (95% CI [−0.17, +0.06]), consistent with width-independence and inside the rule-to-rule range quoted above. What does not tighten is the crossing itself: at $m = 32$ the HAR curve reads 0.5026 ± 0.0524 at $\alpha = 10^{-3}$, so the crossing sits within one seed standard error of a grid node and the interpolated α_c is decided by the draw as much as by the grid. Nine widths therefore sharpen the ordering and the sign of the intermediate exponent without pinning its value, and the main text keeps the four-width estimates for that reason.

Encoder architecture does not change what the probe reads. Repeating the one-probe calibration with the Transformer encoder of Section 6.2 at $m \in \{32, 64, 128\}$, ten seeds and otherwise the protocol of Section 6.1, gives probe exponents $\gamma_{\text{HAR}}^{\text{probe}} = -0.93$ ($R^2 = 0.99$), $\gamma_{\text{bat}}^{\text{probe}} = +0.39$ ($R^2 = 0.87$) and $\gamma_{\text{RML}}^{\text{probe}} = -0.17$ ($R^2 = 0.79$), against -1.63 ($R^2 = 0.96$), $+0.67$ ($R^2 = 0.86$) and $+0.28$ ($R^2 = 0.40$) for the harness MLP. The signs agree on HAR (both negative) and on battery (both positive), RadioML is consistent with zero under both, and the 95% intervals overlap in all three domains: the calibration’s width-dependence is a property of the domain, not of the encoder. This is the capacity-side counterpart of the protection transfer of Section 6.2, and `probe_exponent.py` reproduces it from the shipped calibration runs.

These are probe exponents, not the crossing exponents of Section 7.1. The two estimators agree in sign on RadioML (+0.28 against +0.02) and disagree on HAR (−1.63 against +0.38) and battery (+0.67 against −0.06). The crossing locates the dose at which the *trained* scatter has lost half its variance, whereas the probe reads $1/\hat{\rho}$ on the healthy anchor; the two place the transition within a factor of a few of each other at a fixed width (a factor of 7 at $m=64$ on NYUv2) but need not share its width-scaling. Where they do not, it is the probe that leaves the admissible range: (11) admits only $\gamma \geq 0$, so a negative probe exponent is not a capacity exponent. The exponents of Section 7.1 are therefore the crossing ones, and the probe is reported for NYUv2 (Appendix E) because there it is the quantity STFCAL itself gates on.

Several sweeps show bimodal per-seed scatter near an empirical crossing. This is consistent with multiple outcomes but does not uniquely identify a saddle-node bifurcation; initialization, finite training time and optimization can also generate heterogeneous outcomes.

All numbers use deterministic seeds, bf16 autocast with TF32 matmul on an NVIDIA A100 80 GB, Adam ($\text{lr} = 10^{-3}$, batch 256). The relative-collapse threshold is $0.3\times$ the per-seed single-task ceiling; recovery is defined relative to the same-seed joint baseline. Code, seeds, hyper-parameters and per-fold tables ship with the repository.

C GENERALIZATIONS AND FINITE-SIZE THEORY

Proposition 4 (Fixed attenuation in a specified model). *For $T \succeq 0$ in restoration-normalized coordinates, uniform attenuation $T \mapsto (1 - \lambda)T$, $0 \leq \lambda < 1$, multiplies the first-modal-contraction threshold by $1/(1 - \lambda)$. At fixed dose the smallest λ meeting $\alpha(1 - \lambda)\|T\| \leq 1$ is $\max(0, 1 - 1/(\alpha\|T\|))$ when $\alpha\|T\| > 0$.*

Proof. All eigenvalues scale by $1 - \lambda$; solve the displayed inequality. This statement concerns a fixed PSD model, not arbitrary signed or nonsymmetric filtered dynamics. $\square$

Theorem 4 (Annihilation by orthogonal compression). *Let $T \succeq 0$, $U = \text{range}(T)$, and $Q = I - P_S$ be an orthogonal projector. Then $QTQ = 0$ iff $U \subseteq S$. Among such deletions the unique smallest deleted subspace is U , and $I - P_U$ preserves its orthogonal complement. This is an algebraic statement about two-sided compression.*

Proof. $QTQ = (T^{1/2}Q)^\top (T^{1/2}Q) = 0$ iff $\text{range}(Q) \subseteq \ker T$, equivalently $U \subseteq S$. A general one-sided backward operator QT need not describe symmetric neural-network dynamics; a signed filter can even reverse contraction, so its operator norm cannot supply the missing implication. $\square$

Corollary 3 (Subspace estimation bound). *If $\|P_U - P_{\hat{U}}\| \leq \epsilon$ and $Q = I - P_{\hat{U}}$, then*

$$\|QT\| \leq \epsilon\|T\|, \quad \|QTQ\| \leq \epsilon^2\|T\|. \quad (15)$$

Proof. Use $T = P_U T P_U$ and $\|Q P_U\| \leq \epsilon$ on one or both sides. Whitening must precede this argument if T is restoration-normalized; an angle bound in other coordinates is insufficient. These are upper bounds on residual magnitude, not evidence of a nonzero residual or inevitable failure at a finite dose. $\square$

Theorem 5 (Minimum rank for a PSD compression). *Fix a PSD operator T in specified Euclidean coordinates, with eigenvalues $\kappa_1 \geq \dots \geq \kappa_d \geq 0$, and a threshold $\tau \geq 0$. Among orthogonal projectors $Q = I - P_S$ satisfying $\|QTQ\| \leq \tau$, the minimum deletion rank is*

$$r^* = \#\{j : \kappa_j > \tau\}.$$

Deleting $S_c = \text{span}\{u_j : \kappa_j > \tau\}$ attains it. Within the class of spectral deletions, S_c also minimizes any inclusion-monotone deleted energy. Uniqueness of the energy optimum requires positive energy in every additionally deleted eigendirection.

Proof. The spectral choice leaves only eigenvalues at most τ . For any $\dim S < r^*$, dimension counting gives a unit vector $v \in S^\perp \cap S_c$. Its Rayleigh quotient satisfies $v^\top QTQv = v^\top T v > \tau$, violating feasibility. For spectral deletions feasibility requires containing S_c , giving the energy statement by inclusion. Rotated subspaces can also attain the same minimum rank; no global minimum-energy claim over rotations follows. $\square$

Corollary 4 (Saturation and its limits). *For $\tau = 1/\alpha$, the optimal rank changes in steps as α crosses inverse positive eigenvalues and eventually equals $\dim U$. The limiting projector is $I - P_U$, which equals full detachment only on U , or globally when U is the entire gradient space. Exhausting a finite probe depth does not prove saturation of U .*

Which basis is justified? The theorem concerns a specified operator, not a mean of gradient-norm ratios. For a linear reconstruction decoder Wh with squared loss, $g_a = W^\top \delta$ (up to loss normalization) and the Hessian is $W^\top W$. The uncentered second moment relevant to normalized residual energy is

$$C_g = \mathbb{E} \left[\frac{g_a g_a^\top}{(\|g_p\| + \varepsilon)^2} \right] = W^\top \mathbb{E} \left[\frac{\delta \delta^\top}{(\|g_p\| + \varepsilon)^2} \right] W.$$

For any rank- r projector P , the squared residual is $\text{tr}((I - P)C_g)$, minimized by the top r eigenvectors of C_g (the Rayleigh–Ritz principle). Its square root upper-bounds the corresponding mean norm by Jensen’s inequality. It does not generally minimize that mean itself. Hessian eigenvectors coincide with this basis under isotropic weighted residual covariance; representation PCA requires its leading subspace to coincide with the leading eigenspace of C_g . Commutation gives a common eigenbasis but does not ensure the same ordering. A nonlinear decoder additionally contributes residual-weighted second derivatives to its Hessian. Thus a nearly flat PCA residual curve is consistent with gradient energy outside leading representation components, but does not prove that the reconstruction residual is orthogonal to those components.

The variance-shrinkage rank sweep is reported in Table 4. For real HAR reconstruction the auxiliary basis uses 22.6 versus PCA’s 57.0 directions at $\alpha = 12$, with similar accuracy (Table 5); this supports basis sensitivity, not universal rank optimality. The RMS-basis pilot is retained in the released research records, including its negative accuracy result at three seeds.

MATR audit. The ten-seed cap32 run gives SOH RMSE 5.26 for `stfcalrank` versus 5.24 for joint; its reported rank reaches the cap in every seed. Increasing probe depth to 32 therefore does not validate minimum-rank protection. The single-seed full-depth audit gives across-example covariance traces 3.131, 4.065, 0.906, and 0.821 for joint, detachment, calibrated fallback, and the rank arm, respectively; the SOH-only reference is 1.147. Corresponding normalized SOH prediction variances are 0.00826, 0.01380, 0.01253, and 0.01019, versus reference 0.01379. Representation variance, effective rank and prediction variance must be reported separately: these values neither establish representation recovery nor a constant-prediction collapse. Historical battery checkpoints were selected on their evaluation split; they remain exploratory. A new cell-disjoint validation protocol fits preprocessing on training cells and evaluates test data only after checkpoint selection.

Parameter-space variance constraint. A complementary experimental intervention uses $v = \nabla_{\theta} V$, proposed update $u_0 = G_p + \alpha G_a$, and slack $h = V - V_{\min}$. For $k > 0$ and $v \neq 0$, the closest update satisfying $v^{\top} u \leq kh$ is

$$u^* = u_0 - \frac{(v^{\top} u_0 - kh)_+}{\|v\|^2} v.$$

The KKT conditions prove this Euclidean halfspace projection. Under a well-defined continuous flow, $\dot{h} \geq -kh$ preserves feasibility on this *fixed batch*. At $v = 0$ feasibility requires $kh \geq 0$. Discrete updates require checking the actual post-step variance: the supplied SGD implementation backtracks and restores parameters on failure. The argument covers neither unseen data nor Adam without modeling its applied update. This applies established constrained-gradient ideas (Allibhoy & Cortés, 2024), without claiming halfspace projection as novel; empirical benefits remain to be established.

Table 4: Dose-to-rank curve on UCI-HAR ($m = 64$, five seeds, probe depth $R = 64$): the derived rank r^* grows monotonically with the operating weight, while STF-0 deletes all 64 directions at every dose. Below the crossing the gate is shut and no direction is removed. Relative scatter is S_h/S_{single} for the principal-basis rank arm (stfcalrank); ranks use the two bases.

α	gate	r^* (principal)	r^* (aux. basis)	rel. scatter (rank arm)	rel. scatter (JOINT)
0.003	0	–	–	0.404	0.402
0.01	0	–	–	0.287	0.280
0.03	1	1.6	1.6	1.001	0.200
0.1	1	3.8	3.8	1.026	0.138
0.3	1	4.2	4.2	1.018	0.101
1.0	1	11.6	11.6	0.978	0.070
3.0	1	25.8	25.8	0.987	0.051
10	1	39.8	40.0	0.986	0.035
30	1	47.4	47.8	0.968	0.025
100	1	61.6	61.0	0.950	0.016

Table 5: Real HAR reconstruction head ($m = R = 64$, ten seeds): stfcalrank, means across seeds. Auxiliary-gradient directions reduce deletion at moderate doses with similar accuracy; no uniform dominance is claimed. The separate stfcal fallback is basis-independent. Raw runs: results/XDRBASIS-REAL/.

α	basis	deleted rank	relative scatter	accuracy
12	pca	57.0	0.842	0.9492
12	aux	22.6	0.811	0.9497
30	pca	62.8	0.754	0.9512
30	aux	44.1	0.746	0.9531
100	pca	64.0	0.630	0.9524
100	aux	59.4	0.654	0.9520

Proposition 5 (Sampling fluctuation of a frozen statistic). *For independent identically distributed scalar diagnostics X_i with finite variance σ_X^2 , $\hat{\rho} = B^{-1} \sum_i X_i$ has variance σ_X^2/B and obeys a central limit theorem. Under a normal approximation the probability that the gate $\alpha\hat{\rho} > 1$ opens is $\Phi(\sqrt{B}(\rho - 1/\alpha)/\sigma_X)$. This is not the probability of training collapse. Batch-dependent centering or fitted bases require additional analysis or an independent frozen calibration sample.*

Proof. Use variance additivity and the central limit theorem. For a ratio A/P , the delta-method variance includes $-2\mu_A \text{Cov}(A, P)/\mu_P^3$ as well as the two marginal variance terms. Mean per-example ratios and ratios of means are distinct estimands. $\square$

Proposition 6 (Conditional calibration bound). *Assume independent $X_i \in [0, M]$ with common mean ρ and population variance at most v_0 . Bernstein’s inequality gives*

$$\mathbb{P}(|\hat{\rho} - \rho| \geq \varepsilon) \leq 2 \exp \left[-\frac{B\varepsilon^2}{2v_0 + 2M\varepsilon/3} \right]. \quad (16)$$

Thus an error probability at most δ follows if

$$B \geq (2v_0 + 2M\varepsilon/3)\varepsilon^{-2} \log(2/\delta). \quad (17)$$

Table 6: Empirical frozen-anchor sampling check related to Proposition 6. Healthy HAR anchor ($m = 64$, four detached warm-up epochs), frozen; the shipped probe re-run on 2000 batches of size B per seed, five seeds, cells drawn with replacement. sd and CV are the spread of $\hat{\rho}$ across batches at a fixed anchor; “meas.” is the observed $\mathbb{P}(|\hat{\rho} - \rho| > \eta\rho)$ and “bound” the Bernstein envelope (16) at the same η , evaluated from the probe’s own moments ($\kappa^2 = 2\rho^2 c_0^2 + \frac{2}{3}R_{\max}\epsilon$). The plug-in envelope exceeds observed frequencies in these runs; this comparison does not establish a population confidence guarantee. Fitted slopes: CV vs. B , -0.500 ($R^2 = 0.96$); $\hat{\rho}$ vs. B , $+0.006$; and $\sum_i g_a^{(i)}$ vanishing to 4×10^{-8} of $\sum_i \|g_a^{(i)}\|$. Raising confidence from 90% to 99.9% scales the prescription by $\ln(2/10^{-3})/\ln(2/0.1) = 2.54$.

B	$\hat{\rho}$	sd($\hat{\rho}$)	CV($\hat{\rho}$)	$\eta = 0.2$		$\eta = 0.1$	
				meas.	bound	meas.	bound
16	74.30	69.52	0.947	0.820	1.94	0.911	1.98
32	76.69	50.40	0.664	0.740	1.90	0.866	1.97
64	77.03	36.00	0.471	0.657	1.83	0.836	1.94
128	77.00	25.34	0.332	0.526	1.71	0.756	1.90
256	77.74	18.24	0.237	0.378	1.49	0.657	1.83
512	77.71	12.92	0.168	0.221	1.15	0.538	1.69
1024	77.97	9.20	0.119	0.094	0.70	0.384	1.45
2048	78.00	6.34	0.082	0.022	0.29	0.218	1.07
4096	77.90	4.54	0.059	0.003	0.06	0.095	0.61

A nonzero primary-gradient denominator alone does not provide a uniform bound M . Population bounds cannot be replaced by observed sample moments without a separate valid empirical-concentration result.

Proof. Apply the bounded-variable Bernstein inequality to the centered independent sum and solve for B . $\square$

Corollary 5 (Certifying a frozen gate or nested rank rule). *Under Proposition 6, a gate with population margin $d = |\rho - 1/\alpha| > 0$ has error probability at most $2 \exp[-Bd^2/(2v_0 + 2Md/3)]$. For $R + 1$ fixed residual statistics with uniformly positive margins, a union bound controls all their threshold decisions and hence the selected nested rank. This does not identify the statistics with Hessian eigenvalues, certify an adaptively fitted basis without conditioning, or certify a subsequent training trajectory.*

Proof. A wrong threshold decision requires an estimation error at least its margin; apply the bound and then the union bound. Strict margins exclude equality cases. $\square$

Proposition 7 (Local linear noise floor). *For deviations $e_{t+1} = ae_t + \eta\xi_t$ from a fixed equilibrium, assume $|a| < 1$ and independent mean-zero innovations of variance D . The stationary variance is $\eta^2 D/(1 - a^2)$.*

Proof. The second-moment recursion is $\mathbb{E}e_{t+1}^2 = a^2\mathbb{E}e_t^2 + \eta^2 D$; sum the convergent geometric series. This is variance of deviations, not total representation scatter. Divergence of a local linear approximation as $|a| \rightarrow 1$ does not prove a universal ordering of half-height and binary-collapse thresholds in a nonlinear experiment. $\square$

Repeated frozen-anchor probes provide an empirical sampling check. On HAR, 2000 resampled batches per size and five seeds yield fitted slopes -0.500 for the coefficient of variation and -0.494 for the standard deviation, while the mean is approximately flat (slope $+0.006$). Table 6 retains the numerical plug-in envelopes computed in those runs. Since its variance and range were estimated from the same data, and the implemented batch statistic can share a center across examples, these envelopes are diagnostic comparisons rather than rigorous population certificates. The released probe also distinguishes mean per-example ratios from ratios of batch means; the latter can vanish for a centered variance loss. These empirical facts do not justify exchanging the two estimands.

Table 7: The four scientific domains and the financial discovery domain, with their primary and auxiliary structure and evaluation metrics. Auxiliary toxicity is instantiated identically everywhere (variance-shrinkage projections); the flagship battery application and the NYUv2 dense benchmark (Appendix E) also pair the encoder with practical reconstruction heads.

Domain	Signal	Primary task	Samples	Metric
HAR (UCI-HAR)	motion (561-d)	6-class activity	7,352 / 2,947	accuracy
RadioML (2016.10A)	I/Q (256-d)	11-class modulation	110,000 / 110,000	accuracy
Battery (NASA+MATR)	charge/discharge	SOH + RUL	18,696 / 3,210	RMSE / MAE
Finance (A-share)	1,680-d factors	5-d return rank	800 stocks $\times$ 2,059 d	rank-IC
NYUv2	RGB-D (144 $\times$ 192)	13-class seg + depth + normals	795 / 654 scenes	pxacc / mIoU / RMSE / angle

C.1 PREDICTION VARIANCE NORMALIZATION AND TARGET z -SCORING

A linear prediction head $\hat{y} = w^\top h + b$ introduces a scale ambiguity: shrinking centered representations by $1/c$ and expanding w by c preserves predictions. Shrinking both generally does not preserve the regression loss, nor can constant predictions fit varying targets exactly. Consequently prediction variance and across-example representation covariance should both accompany task error. Prediction normalization and target scaling are practical controls, not proofs of a universal prediction-level trap.

The historical regression recipe standardizes targets and divides predictions by a detached batch standard deviation. This changes optimization and can reduce sensitivity to scale, but detached normalization does not implement the derivative of a truly scale-invariant loss. Softmax also does not normalize logit scale: changing that scale changes probabilities. These operations therefore require empirical ablations, with scaling statistics fit only on training data.

D ADDITIONAL TABLES AND PROTOCOL

Algorithm 1 STF training protocol (per domain, per configuration)

- 1: **Input:** domain data $\mathcal{D}$, encoder width m , toxicity schedule $\{\alpha\}$, seeds
- 2: **Phase 1 – health anchor:** train the single-task encoder ($\alpha = 0$), record healthy scatter $\mathcal{S}_{single} = \mathcal{S}_h(\theta^*)$ and healthy spectral ratio ρ_0 .
- 3: **Phase 2 – calibration:** sweep α with mode JOINT; record ρ , relative scatter $\mathcal{S}_h/\mathcal{S}_{single}$, collapse flag; locate α_c by the threshold crossing.
- 4: **Phase 3 – repair:** train JOINT, STF-0, STF-HARD, STF-SOFT at supercritical $\alpha > \alpha_c$; per step: forward $h = E_\theta(x)$ and form $L = L_p + \alpha L_a$; backprop to obtain g_p and $g_a^{(k)}$; update $\theta \leftarrow \theta - \eta J_\theta^\top (g_p + \sum_k F_\beta[g_a^{(k)}])$, $J_\theta = \partial h / \partial \theta$; monitor ρ and $\mathcal{S}_h/\mathcal{S}_{single}$.
- 5: **Report:** collapse rate, recovery rate, ρ , $\mathcal{S}_h/\mathcal{S}_{single}$.

On NASA, the historical batch-256 ten-seed replication gives joint SOH RMSE 11.95 ± 0.79 , detachment 9.33 ± 0.50 , and calibrated filtering 11.65 ± 0.97 , against the single-task reference 9.20. The probe reads 0.29 and leaves the gate closed. This is a false negative relative to the observed task benefit of detachment; it does not isolate a prediction-level mechanism. The historical scatter is coordinate variance, and checkpoint selection uses the evaluation split.

The historical ten-seed MATR calibrated-fallback row improves SOH over joint (1.77 ± 0.47 versus 5.24 ± 0.54), while full detachment reaches 1.68 ± 0.41 . Censored RUL requires separate observed-error and censored-shortfall metrics; proximity to a particular collapsed MAE is not a valid quality ordering. At cap32 the rank arm remains near joint (5.26 SOH RMSE), so the fallback result does not validate minimal-rank protection. NASA’s closed gate and the discrepancy from detachment are empirical limitations of the diagnostic. All these historical checkpoints were chosen using the evaluation split, motivating the independent validation protocol described above.

A single-term attribution on MATR (4 seeds) isolates the gate factors: the fast-charge gain is carried by the rate term alone (fast-charge SOH-RMSE 0.95 without it) and the extreme-temperature gain by the temperature term alone (2.35 without it), confirming that the percentile thresholds select the intended operating windows rather than an artifact of the joint gate; the ω -sweep over $\{0.5, 1, 2\}$ brackets the operating point at $\omega=1$.

Algorithm 2 Calibrated spectral filtering (STFCAL); no tuned hyper-parameter beyond the operating weight α already chosen by the practitioner

- 1: **Input:** data $\mathcal{D}$, encoder width m , operating weight α , probe depth R , warm-up length W
- 2: **Warm-up:** train θ with the auxiliary branch detached ($\nabla_{\theta} L_a \equiv 0$) for W epochs; record the healthy anchor $\mathcal{S}^*$ of (10)
- 3: **Probe:** on one batch of B cells with per-cell gradients $g_p^{(i)}, g_a^{(i)}$ and centered representation H_c with principal basis U , average the per-cell ratio: $\hat{\rho} = B^{-1} \sum_i \|\Pi_c g_a^{(i)}\| / \|g_p^{(i)}\|$ and $\hat{\rho}_{\text{res}}[r] = B^{-1} \sum_i \|\Pi_c (I - \sum_{j \leq r} u_j u_j^\top) g_a^{(i)}\| / \|g_p^{(i)}\|$ for $r = 0, \dots, R$ (the batch-mean ratio $\|\Pi_c \bar{g}_a\| / \|\bar{g}_p\|$ vanishes for a variance-shrinking auxiliary)
- 4: **Calibrate:** gate $\leftarrow 1[\alpha \hat{\rho} \geq 1]$; **if** gate = 1 **then** $\mathcal{R} \leftarrow \{r \leq R : \alpha \hat{\rho}_{\text{res}}[r] < 1\}$; **if** $\mathcal{R} = \emptyset$ **then** $r \leftarrow \dim H_c$ (no safe rank within the probe: fall back to full deletion) **else** $r \leftarrow \min \mathcal{R}$ **else** $r \leftarrow 0$
- 5: **Latch:** freeze (r, U) for the rest of training; apply $F[g_a] = g_a - \sum_{j \leq r} (u_j^\top g_a) u_j$ to the parameter update through $J_\theta = \partial h / \partial \theta$ every step
- 6: **Report:** gate, $r, \hat{\rho}, \mathcal{S}_h / \mathcal{S}^*$

Table 8: Repair on the discovery domain (A-share ranking, 10-fold walk-forward $\times 10$ seeds, OOS rank-IC, mean $\pm$ std over 100 fold-cells). STF-0 *deletes* the auxiliary signal (−39% versus joint, paired $p=5 \times 10^{-5}$); both spectral gates retain it (STF-HARD +5%, STF-SOFT −7%, neither significant), as predicted by Corollary 1.

Mode	OOS rank-IC	vs. joint
JOINT	+0.060 $\pm$ 0.055	—
STF-0 (.detach())	+0.036 $\pm$ 0.056	−39%
STF-HARD	+0.063 $\pm$ 0.052	+5%
STF-SOFT	+0.056 $\pm$ 0.053	−7%

The full grid behind the two predictions of Section 6.5 is Table 18. On HAR the conflict operator changes accuracy by < 0.010 once the filter has acted (0.943 alone versus 0.933/0.940 composed), and the unfiltered representation never leaves the collapsed band (relative scatter ≤ 0.15 against the filtered 0.97–1.14).

The synthetic testbed uses a 17-point log-spaced α grid over six widths with untrained random projections of rank $\lfloor m/2 \rfloor$; its toxic rank is a fixed fraction of the width, so it realizes $\beta = 1$ and yields $\gamma = 0.07$ under the relative half-height rule ($\gamma = 1.53$ under the absolute rule, the artifact quantified above), the dilution predicted by Proposition 3. The RadioML subcritical sweep ($\alpha \in [3 \times 10^{-3}, 1]$, 5 seeds, densified about the transition) locates crossings $\alpha_c = \{0.17, 0.39, 0.24, 0.22\}$ for $m = \{32, 64, 128, 256\}$: the eightfold width range moves the crossing by a factor of 2.3, against the eightfold of mean-field $\gamma = 1$ and more than an order of magnitude for the undiluted $\gamma_0 = 2.43$. The crossing is not resolved more finely than that by this grid — at $m=64$ the per-seed crossings alone span 0.29–0.54, so the width trend sits inside the seed spread — and the near-zero R^2 of the power-law fit is that same flatness read off the line.

The accompanying repository contains everything needed to regenerate every number in this paper. The cross-domain protocol (`cross_domain_mtl.py`), the flagship battery application (`battery_bmtl_v3.py`), and the NYUv2 dense benchmark (`nyu_bmtl.py`) implement the whole STF family, including the calibrated variants STFCAL and STFCALRANK and the single-probe calibration of Algorithm 2; the reported conflict-resolution baselines (PCGrad, CAGrad, FAMO) share the same harness, and so does Table 18. The remaining families (MGDA, Aligned-MTL, Nash-MTL, IMTL-G, IMTL-L, GradNorm, uncertainty weighting) are benchmarked through it: `moocombiners.py` ports the kernels of the reference releases of the original authors and `cross_domain_mtl.py` combines them in one code path, so the collapse rates of Table 15 and the battery numbers of Table 10 are measured under the same optimizer, schedule and batch as the baselines they extend, with each family at its published defaults.

Each experiment is specified as a JSON job manifest — one entry per (domain, width, mode, seed) cell, carrying the exact command line, output path and log path — and executed by a deterministic runner that skips cells whose outputs already exist; the manifests cover the four-domain collapse scans, the width-resolved capacity sweeps, the conflict-composition grid, the real-auxiliary-

Table 9: Censoring-supervision ablation on MATR, in SOH RMSE% and RUL MAE (cycles; leave-cell-out, $d=128$; the legacy rows use 4 seeds and the censoring-aware row 10; bf16 training / fp32 evaluation). *Legacy* treats right-censored lower bounds as exact targets (and, in the eol-only variant, discards the 96/185 cells that never reach EOL, leaving a 14-cell test split); *censoring-aware* applies the Tobit-style one-sided hinge to the 59.2% censored samples and retains all cells.

Mode	SOH rms(%)		RUL MAE (cyc)		cells used
	JOINT	STF-0	JOINT	STF-0	
legacy (RUL=0 labels)	5.56 $\pm$ 0.17	2.05 $\pm$ 0.38	94.1 $\pm$ 5.7	97.0 $\pm$ 15.3	185
legacy + --eol-only	1.53 $\pm$ 0.13	0.53 $\pm$ 0.03	77.5 $\pm$ 2.3	76.6 $\pm$ 4.1	89
censoring-aware (ours)	5.24 $\pm$ 0.54	1.68 $\pm$ 0.41	209.0 $\pm$ 59.8	574.7 $\pm$ 80.7	185

legacy rows reproduce the nocens supervision, every other setting held fixed.

Table 10: Numeric multi-task baselines on the real battery path at supercritical $\alpha=10$ (5 seeds), reported as SOH-RMSE% $\downarrow$, RUL-MAE in cycles $\downarrow$, and raw representation scatter $\mathcal{S}_h$ (with the ceiling scatter of Table 2 for scale; the collapse rule of Appendix D applies to the relative scatter $\mathcal{S}_h/\mathcal{S}_{\text{single}}$, not to these raw values). Replacing plain summation with PCGrad, FAMO, or CAGrad leaves the battery encoder degraded at the flagship caliber: on MATR every combiner sits at 4.0–10.1% SOH against 1.83% for STF-0, and on NASA at 11.9–16.0% against 8.81%. The widest raw scatter among the combiners reflects a reweighted collapse, not a healthy encoder: every one of them remains collapsed (1.0 rate) in Table 15. Single-task ceilings are those of Table 2, whose STF-0 row (1.68 $\pm$ 0.41) is the flagship ten-seed configuration; the row here is re-measured at this dose with five seeds, and every STF-0 value above lies inside the corresponding ten-seed interval of that row (1.83 $\pm$ 0.51 vs. 1.68 $\pm$ 0.41 SOH on MATR, 8.81 $\pm$ 0.72 vs. 9.35 $\pm$ 0.55 on NASA, and both RUL values within the ten-seed $\pm$), so the two runs agree within seed noise.

Combiner	MATR			NASA		
	SOH rms(%)	RUL MAE	scatter	SOH rms(%)	RUL MAE	scatter
single-task reference	0.51	694.6	–	9.17	18.5	–
JOINT (sum, plain)	6.43 $\pm$ 0.19	208.8	0.126	13.33 $\pm$ 1.07	18.7	0.214
PCGrad	4.02 $\pm$ 0.98	172.9	0.094	11.89 $\pm$ 0.73	18.6	0.164
FAMO	10.10 $\pm$ 2.17	295.3	0.435	16.02 $\pm$ 0.50	32.4	0.474
CAGrad	6.33 $\pm$ 0.45	198.0	0.166	13.04 $\pm$ 0.75	17.7	0.198
STF-0 (ours)	1.83 $\pm$ 0.51	623.4	0.198	8.81 $\pm$ 0.72	17.1	0.225

head comparisons, the auxiliary-weight sweep of Table 11, the calibrated validation of Section 5.1, the NYUv2 calibration, dose–response, and dense benchmark of Appendix E, and the multi-task-baseline coverage of Table 15 in three domains. Every reported cell uses fixed integer seeds (4–10 per cell), each table states its seed count, and the JSON outputs retain per-seed values.

All runs use one NVIDIA A100 80 GB; Adam at $\text{lr} = 10^{-3}$; bf16 autocast with TF32 matmul; the calibration probe is evaluated in fp32 outside autocast on a detached clone. Batch size is 256 for the unified protocol, the real-auxiliary comparison, the regularizer sweep and the MATR flagship runs, 128 for the NASA conflict baselines (256 for its plain-summation runs), 32 for NYUv2, and 512 for the walk-forward discovery domain. A unified-protocol cell is a few GPU-minutes (the HAR calibration of Section 5.1 runs ten seeds in a few minutes), while the flagship battery runs of Section 6.3 take 150 epochs per configuration. The finite-batch claims of Proposition 6 are directly checkable from the artifacts: the probe persists $\hat{\rho}$, the residual family $\hat{\rho}_{\text{res}}[0..R]$, and the gate and rank it induces, for every seed, and `calib_cert_b.py` re-probes a frozen healthy anchor over a sweep of probe sizes, which is Table 6.

The cost of the filter relative to the alternatives is measured directly. Table 12 times one training epoch on UCI-HAR at four widths for every variant of the family and for the gradient-surgery baselines, reporting the median over ten repetitions of the per-repetition median epoch time, with epoch 0 dropped as warm-up and a device synchronisation inside each timed epoch. Two guards keep the ratio honest. The run starts only after the device reports idle for three consecutive samples, and the row order is rotated by one position per repetition, so no method is always measured first or last. Both matter: the same script, run while a co-tenant held the device, reported the reference epoch at 0.180 s instead of 0.099 s and FAMO at 0.0% instead of +31%, because the row that is always measured last absorbs whatever drift the device has. On the idle device the spectral variants cost

Table 11: Auxiliary-weight (dose-response) sensitivity on MATR, in SOH RMSE (%) and RUL MAE (cycles) at seed 0, with censoring-aware supervision and bf16 training / fp32 evaluation. JOINT trains the auxiliary reconstruction/distributional heads with loss weight λ_{aux} ; STF-0 detaches them, so λ_{aux} never enters its computation and its row is identical at every dose. SOH RMSE (%) degrades with the dose for JOINT at every width, while the benign dose $\lambda_{\text{aux}}=10^{-2}$ recovers near-STF-0 SOH. Single-task ceilings are the SOH-only model (SOH columns) and the RUL-only model (RUL columns) at each width.

Mode	λ_{aux}	SOH RMSE (%)↓			RUL MAE (cyc)↓		
		$d=32$	$d=128$	$d=256$	$d=32$	$d=128$	$d=256$
JOINT	10^{-2}	2.72	1.36	1.52	478.0	386.3	377.5
	10^{-1}	2.69	2.23	2.87	230.4	399.7	411.0
	1	5.96	4.69	5.53	285.6	198.0	176.4
	10	6.52	6.58	6.01	178.5	197.0	182.0
STF-0 (dose-free)	—	2.36	1.18	1.51	607.2	627.4	540.2
single-task ceiling	—	0.60	0.33	0.57	951.8	761.7	325.8

Table 12: Per-epoch wall-clock cost of the STF family and of the gradient-surgery baselines on UCI-HAR at four widths (batch 256), as a percentage of the unfiltered plain-summation step. Each cell is the median over ten repetitions of the per-repetition median epoch time, with epoch 0 dropped as warm-up and a device synchronisation inside each timed epoch. The row order is rotated by one position per repetition and the run starts only after the device reports idle for three consecutive samples, so neither warm-up nor drift is charged to a method; an estimator that pairs the ratio inside each rotation position agrees with every cell to within 5 percentage points. Plain summation takes 0.0813/0.0986/0.1283/0.1454 s per epoch at $m=32/64/128/256$. The detaching variants are cheaper than the unfiltered step because the toxic head no longer back-propagates through the encoder; the spectral variants are the cheapest non-detaching option at every width, and their relative cost is largest at the smallest width because the projection is $O(Bd)$ against the encoder’s $O(Bd^2)$.

Method	$m=32$	$m=64$	$m=128$	$m=256$
plain summation (s/epoch)	0.0813	0.0986	0.1283	0.1454
<i>overhead relative to plain summation</i>				
STF-0 (detach)	−35.8%	−48.1%	−59.5%	−65.0%
STF-HARD	+21.8%	+18.1%	+14.3%	+9.5%
STF-SOFT	+21.8%	+19.8%	+8.9%	+12.2%
STFCAL	−0.7%	−20.1%	−37.7%	−47.8%
PCGrad	+41.6%	+31.4%	+26.6%	+21.1%
CAGrad	+75.2%	+60.2%	+46.0%	+38.5%
FAMO	+31.0%	+31.1%	+23.9%	+19.6%

21.8% at $m=32$, falling to 9.5% (STF-HARD) and 12.2% (STF-SOFT) at $m=256$: the projection is $O(Bd)$ against the encoder’s $O(Bd^2)$, so its relative cost shrinks as the encoder grows. The detaching variants are *cheaper* than the unfiltered step at every width, by up to 65.0% (STF-0) and 47.8% (STFCAL), because `.detach()` and the calibrated gate remove the toxic head’s backward pass through the encoder. Gradient surgery is dearer at every width — PCGrad 41.6% to 21.1%, CAGrad 75.2% to 38.5%, the price of a second per-task backward pass through the shared encoder — and FAMO’s learned scalarisation is not free either, at 31.0% to 19.6%. The spectral variants are therefore the cheapest non-detaching option at every width, and no STF variant exceeds 21.8%. The same matrix, repeated on the other four domains at the same four widths, keeps the ordering: over the twenty cells the spectral variants cost +6.3% to +26.0% against +5.8% to +98.2% for the surgery baselines, and both are cheaper than PCGrad and CAGrad in every cell. The single exception to the full ordering is FAMO at $m=256$ on CelebA and NYUv2, where it comes within six points of the cheaper spectral variant.

E NYUV2: DENSE PERCEPTION BEYOND TIME SERIES

NYUV2 (Silberman et al., 2012; Ladický et al., 2014) provides 795 training and 654 test scenes with per-pixel semantic segmentation (13 valid classes; boundary pixels labeled -1 and masked in loss and metric), monocular depth, and surface normals; we downsample to 144×192 . The benchmark encoder is convolutional (four stride-2 blocks with batch-norm-ReLU, $3 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ channels, flattened to $d=128$) with linear heads for the three tasks; losses are cross-entropy (ignore index -1), L_1 depth, and a cosine normal loss with zero-norm pixels masked; metrics are pixel accuracy, mIoU, depth RMSE/MAE, and mean angular error, all read at the final epoch over ten seeds with no test-set selection. Two auxiliary branches are studied: *proj*, the controlled variance-shrinkage toxifier of the unified protocol over a fixed random projection to $d/4$, at the calibrated supercritical dose $\alpha=1$; and *ae*, a trainable three-layer head reconstructing the input — the production-style reconstruction aux of the battery flagship — at $\alpha=250$, $3.4\times$ its calibrated α_c . In the unified-protocol variant the primary task is the 9×12 coarse depth grid (108 outputs) and the harness MLP, optimizer, probe, and collapse criterion are exactly those of Section 6.1.

One probe per seed after four detached warm-up epochs on the harness encoder gives $\hat{\rho} = 2.26 \pm 0.16$, 2.01 ± 0.10 , 1.64 ± 0.09 , 1.29 ± 0.05 at $m=32, 64, 128, 256$ (per-seed ranges 2.06–2.65, 1.90–2.20, 1.51–1.77, 1.22–1.38), i.e. $\alpha_c = 1/\hat{\rho} = 0.44/0.50/0.61/0.78$: the critical dose *grows* with width, and the four-point log-log fit gives capacity exponent $\gamma = 0.27$ ($R^2 = 0.98$, 95% CI $[+0.15, +0.40]$), inside the HAR–RadioML range and consistent with Proposition 3. The estimator differs from the one used for the other domains, and the difference is not cosmetic: here α_c comes from the one-probe calibration ($1/\hat{\rho}$) at four widths, whereas the HAR, battery, and RadioML exponents are read off the scatter half-height crossing. That crossing, applied here, returns the *opposite* ordering — $\gamma = -0.77$ over the same four widths — and the inversion is not an artefact of the threshold that reads it: sweeping the crossing level from 0.3 to 0.7 moves the exponent only between -0.94 and -0.73 ($R^2 \geq 0.99$), whereas the same sweep on HAR stays positive throughout ($+0.34$ to $+0.46$). The two estimators are therefore not interchangeable on this domain: they place the transition within a factor of 7 of each other at $m=64$ (0.069 against 0.50) but disagree on how it moves with width. We report the probe because it is the quantity STFCAL gates on, and quote the crossing’s value beside it so the two can be compared directly. The crossing is not a grid artefact — extending the sweep four decades lower puts the scatter back on the anchor (0.99–1.01 at $\alpha = 10^{-6}$, against 0.64–0.83 at $\alpha = 0.01$, the shallowest dose previously swept) and leaves the crossing where it was — so the inversion is a property of the scatter curve, not of where the grid starts. $\gamma_{\text{NYU}} = 0.27$ is therefore the probe-based value and is not directly comparable with the crossing-based exponents of Section 7.1. The remaining degree of freedom is *when* the probe is read; Appendix F.3 measures it and finds the probe exponent stable in the read time here (it ranges over 0.17, against 2.55 on HAR, which changes sign), and that stability is what makes the probe usable as the estimator on this domain. On the benchmark encoder the same probe reads $\hat{\rho} = 3.27 \pm 0.54$ (per-seed 2.45–3.92), predicting $\alpha_c \approx 0.31$; the *ae* head reads $\hat{\rho} = 0.0136 \pm 0.0015$, predicting $\alpha_c \approx 73$.

Table 13: joint collapses 10/10 for every $\alpha \geq 0.5$ while STFCAL collapses 0/10 with the gate open in every seed (rank 3) and relative scatter 1.00; at $\alpha=0.5$, exactly the predicted shell ($1/\alpha = 2.0$ against per-seed $\hat{\rho} \in [1.90, 2.20]$), the gate opens in exactly the five seeds with $\hat{\rho} > 2$ and every gate-open seed is protected (rank 1 suffices) — an empirical threshold comparison, subject to the scope restrictions of Corollary 5. At sub-critical doses STFCAL is indistinguishable from joint. The collapse is metric-silent: joint depth RMSE *improves* monotonically from 1.335 to 1.092 (-18%) as the relative scatter falls from 0.62 to 0.04, so the supercritical joint configuration *beats* every repaired one on the primary metric while its representation has collapsed to 4% of healthy scatter — the operational case for monitoring ρ rather than the loss (cf. the MATR case in Section 6.2). Wilcoxon on per-seed relative scatter (joint vs. STFCAL) gives $p = 2 \times 10^{-3}$ at every supercritical α , the smallest two-sided value attainable at $n=10$.

Table 14 compares the dense tasks within a shared small-backbone protocol; its absolute metrics are not competitive NYUV2 benchmark claims. The projection auxiliary reduces relative scatter to 0.22 while worsening segmentation and improving depth RMSE, and calibrated filtering restores scatter near the auxiliary-free reference. With the reconstruction auxiliary at dose 250, filtering gives only partial recovery (scatter 3.84 versus reference 7.19). Basis drift and residual gradient energy are possible explanations, but no measured principal-angle trajectory establishes which mechanism dominates. These results motivate directly comparing frozen and refreshed gradient-energy bases.

Table 13: NYUv2 dose-response under the unified protocol, reporting depth RMSE in meters on the 9×12 grid and relative scatter against the per-seed single-task anchor (harness MLP $m=64$, ten seeds; gate and rank from the one-probe calibration at $\hat{\rho} = 2.01$, $\alpha_c = 0.50$, i.e. rank 1 at $\alpha \leq 0.5$ and rank 3 at $\alpha \geq 1$). STF-0 is aux-free at every dose (RMSE 1.353, relative scatter 1.000 at every α).

α	0.03	0.1	0.3	0.5	1	3	10	30	100
JOINT RMSE	1.335	1.336	1.316	1.307	1.305	1.280	1.221	1.100	1.092
JOINT rel. scatter	0.62	0.45	0.30	0.26	0.22	0.16	0.10	0.04	0.04
JOINT collapsed	0/10	0/10	5/10	10/10	10/10	10/10	10/10	10/10	10/10
STFCAL RMSE	1.339	1.327	1.314	1.328	1.354	1.353	1.353	1.354	1.354
STFCAL rel. scatter	0.63	0.45	0.31	0.63	1.01	1.01	1.01	1.00	1.01
STFCAL collapsed	0/10	0/10	4/10	5/10	0/10	0/10	0/10	0/10	0/10
calibrated gate open	0/10	0/10	0/10	5/10	10/10	10/10	10/10	10/10	10/10

Table 14: NYUv2 three-task benchmark on a convolutional encoder ($d=128$, ten seeds, final epoch). Upper block: single-task ceilings and the aux-free three-task baseline. Middle: the controlled *proj* toxifier at the calibrated supercritical dose $\alpha=1$. Lower: the production-style *ae* reconstruction aux at $\alpha=250$. Normal angle in radians (lower is better); depth RMSE in meters; scatter is the cross-sample representation variance. Absolute values are not comparable to published NYUv2 results (small fixed backbone); all rows share the backbone, split, and seeds, so the comparison is within-table.

Configuration	seg pxacc	seg mIoU	depth RMSE	normal angle	scatter
single-seg	0.359	0.083	2.780	1.574	2.22
single-depth	0.076	0.032	1.159	1.573	5.08
single-normal	0.074	0.031	3.346	0.593	12.43
joint3 (no aux)	0.356	0.100	1.130	0.586	7.30
+ proj aux ($\alpha=1$)	0.365	0.090	1.085	0.580	1.64
STF-0	0.356	0.100	1.131	0.585	7.30
STFCAL	0.357	0.100	1.134	0.586	7.37
STFCALRANK	0.357	0.095	1.115	0.583	2.65
+ ae aux ($\alpha=250$)	0.350	0.092	1.168	0.580	3.99
STF-0	0.357	0.099	1.133	0.586	7.19
STFCAL	0.338	0.091	1.171	0.594	3.84

F MATERIAL MOVED FROM THE MAIN TEXT

Per-domain values at ten seeds per cell confirm the calibrated validation of Section 5.1. Supercritical dose — joint relative scatter / STFCAL relative scatter, with collapse rates in parentheses: HAR $\alpha=100$, 0.016 (10/10) vs. 0.990 (0/10); battery $\alpha=100$, 0.147 (10/10) vs. 1.001 (0/10); RadioML $\alpha=3000$, 0.014 (10/10) vs. 0.982 (0/10). Sub-critical dose — STFCAL with the gate closed is indistinguishable from joint: HAR $\alpha=0.001$, 0.525 vs. 0.528; battery $\alpha=0.01$, 0.963 vs. 0.965. Transformer encoder: HAR $\alpha=100$, joint 10/10 collapsed (scatter 0.030), STFCAL 0/10 (1.041); battery $\alpha=100$, joint 10/10 (0.036, SOH-RMSE 0.447), STFCAL 0/10 (1.000, 0.267). CelebA $\alpha=100$ (five seeds): joint 5/5 (0.003), STFCAL 0/5 (0.934).

Table 3 reports the real-auxiliary comparison at $m = 64$ and ten seeds. The two reconstruction heads are benign (no collapse; filtering them is harmless-to-mildly-harmful, as the near-zero measured ρ predicts). The subject probe is a genuine hazard: joint training leaks identity into the representation (probe accuracy 0.646 against 1/21 chance) at a primary-task cost, and the filter restores the primary task while suppressing the probe to chance (0.055). The SNR head at supercritical α enters the saturation regime: joint collapses in every seed, STF-HARD and STF-SOFT cannot recover it, and full deletion performs better in the tested comparison. Measured ρ separates toxic from benign heads by orders of magnitude in every domain — the deployment-time decision the protocol of Section 7.1 calls for.

Table 15: Every gradient-manipulation family of Table 16, benchmarked in the same harness as plain summation at each domain’s supercritical dose (ten seeds per cell). A cell is the collapse rate over seeds and, in parentheses, the mean primary metric — so the HAR column reads accuracy, the battery column log-SOH RMSE, and the RadioML column accuracy. Every family retains the 1.0 collapse rate of JOINT in every domain, and on HAR seven of the ten fall below the 0.945 of JOINT (four to 0.18–0.42): these are outcomes at the specified defaults, not a structural impossibility theorem. Nine of the ten families sit exactly at the 1/11 chance level of RadioML (0.091), the signature of a constant prediction, while JOINT (0.256) and GradNorm (0.170) keep partial signal and still meet the collapse criterion. CAGrad’s $c=0.4$ and $c=0.8$ variants, run on the battery path only, behave identically (collapse rate 1.00; log-SOH RMSE 0.608 and 0.601). STF-0 is the only row that leaves the attractor; its repaired metrics are plotted in Figure 1(c,d) and its recovery rates are in Table 17.

Combiner	HAR acc. $\uparrow$	Battery log-SOH RMSE $\downarrow$	RadioML acc. $\uparrow$
JOINT (plain summation)	1.00 (0.945)	1.00 (0.611)	1.00 (0.256)
<i>conflict resolution</i>			
PCGrad	1.00 (0.585)	1.00 (0.601)	1.00 (0.091)
CAGrad ($c=0.5$)	1.00 (0.952)	1.00 (0.607)	1.00 (0.091)
<i>Pareto</i>			
MGDA	1.00 (0.200)	1.00 (0.385)	1.00 (0.091)
<i>gradient geometry</i>			
Aligned-MTL	1.00 (0.865)	1.00 (0.609)	1.00 (0.091)
IMTL-G	1.00 (0.419)	1.00 (0.601)	1.00 (0.091)
Nash-MTL	1.00 (0.280)	1.00 (0.552)	1.00 (0.091)
<i>loss weighting</i>			
FAMO	1.00 (0.182)	1.00 (0.613)	1.00 (0.091)
UW (uncert.)	1.00 (0.940)	1.00 (0.611)	1.00 (0.091)
GradNorm	1.00 (0.958)	1.00 (0.611)	1.00 (0.170)
IMTL-L	1.00 (0.947)	1.00 (0.611)	1.00 (0.091)
STF-0 (ours)	0.0	0.0	0.0

The experiments delimit where the diagnostics and filters work. Some task metrics improve despite reduced scatter, some real heads are benign at the tested dose, and RadioML has false negatives. A latched warm-up basis can drift, but the subspace-error bound alone does not identify that drift as the cause of an observed failure. Resampling a frozen probe examines sampling variation; it does not certify future training states. Numerical instability and loss of plasticity (Dohare et al., 2024) are further possible failure mechanisms, outside the fixed-operator analysis.

A shared temporal encoder maps each cell’s per-cycle window $x \in \mathbb{R}^{T \times D}$ (normalized voltage, current, temperature, resistance) to a representation $h \in \mathbb{R}^d$; the encoder stacks a TSMIXER (Ekambaram et al., 2023) feature/temporal-mixing block and a gated state-space block (Gu et al., 2022; Gu & Dao, 2023) for long-range degradation context. Three heads share h : a SOH regression head (L_2), a RUL regression head (L_1), and the auxiliary reconstruction/distributional heads of Definition 1.

RUL targets are right-censored, which governs every number below. A cell that has not crossed its end-of-life threshold when recording stops carries only a *lower bound* on its remaining life, so writing the observation depth in as an exact target injects the contradiction “RUL = 0 while SOH $\approx$ 90%” and teaches the model to predict zero life for healthy cells. The problem is severe: 59.2% of MATR samples (96/185 cells) never reach EOL within their window, and MATR summary arrays open with an initialization zero, so a naive threshold test latches onto index 0 for a further 48/185 cells and assigns RUL = 0 to *their entire lifetime*. Discarding them would leave

Table 16: Intervention families and scope. None of the tested neural-network implementations has a universal trajectory guarantee from the projected norm ratio alone. Historical performance comparisons appear in Tables 15 and 19.

Family	Intervention	Relevant limitation
Task weighting (Chen et al., 2018; Liu et al., 2023)	scalar auxiliary strength	positive attenuation can help at a fixed dose
Gradient geometry (Yu et al., 2020; Liu et al., 2021a; Navon et al., 2022)	task-gradient combination	alignment does not determine variance change
Variance regularization (Bardes et al., 2022)	representation penalty	penalty weight and primary-task trade-off
Full detachment	remove auxiliary encoder gradient	also removes useful auxiliary learning
Selective filtering	remove selected directions	basis identification and subsequent drift
Parameter variance constraint	constrain the actual parameter update	fixed guard batch; finite-step feasibility must be checked

Table 17: Collapse rate and recovery rate under supercritical toxicity in three domains (HAR 5 seeds, battery 20 seeds, RadioML 10 seeds). Recovery is the fraction of seeds whose primary metric improves over joint training. On HAR the classifier separates activities even on a collapsed representation, so metric recovery is near-zero there while every STF mode restores the representation; recovery is 1.0 in the domains where the metric itself collapses.

Domain	Metric	JOINT coll. / rec.	STF-0 coll. / rec.	STF-HARD coll. / rec.	STF-SOFT coll. / rec.
HAR	accuracy (flat)	1.0 / 0.0	0.0 / 0.1	0.0 / 0.2	0.0 / 0.1
RadioML	accuracy	1.0 / 0.0	0.0 / 1.0	0.0 / 1.0	0.0 / 1.0
Battery	log-SOH RMSE	1.0 / 0.0	0.0 / 1.0	0.0 / 1.0	0.0 / 1.0

a 14-cell test split, so we supervise the censored cells with the classical Tobit-style form (Tobin, 1958): $|\hat{r} - r|$ for an exact observation and the one-sided hinge $\max(0, r_{lb} - \hat{r})$ for a censored one, penalising only predictions of *less* remaining life than demonstrated. Keeping the EOL search within the physically valid suffix of each trajectory recovers the 48 zero-prefixed cells rather than dropping them (Section 6.4).

Battery degradation is non-stationary: fast-charging and extreme-temperature windows carry disproportionate failure information yet are underrepresented. Following the multiplicative-gate principle (product beats sum under low signal-to-noise), BatteryMTL up-weights the critical windows by

$$\begin{aligned}
 w_i &= w_i^{(\text{rate})} \cdot w_i^{(\text{temp})} \cdot w_i^{(\text{degrad})}, \\
 w_i^{(\text{rate})} &= 1 + \omega \mathbf{1}[\text{C-rate}_i > r_q], \\
 w_i^{(\text{temp})} &= 1 + \omega \mathbf{1}[T_i < t_{lo} \vee T_i > t_{hi}], \\
 w_i^{(\text{degrad})} &= 1 + \omega \frac{|\Delta\text{SOH}_i|}{\max_j |\Delta\text{SOH}_j|},
 \end{aligned} \tag{18}$$

Equation (18) is a soft AND over rate, temperature, and degradation rate. Its thresholds are *dataset-calibrated*: r_q is the 70th percentile of the training-fold C-rate and t_{lo}, t_{hi} its 10th and 90th temperature percentiles. Fixed physical cuts cannot serve here. MATR is a high-throughput fast-charge campaign in which 99.8% of cycles exceed 1.5C, so a 1.5C cut marks almost every sample critical and erases the contrast the gate exists to create, and its 31–36°C band never crosses a 0/40°C rule. Percentile thresholds instead preserve *which* windows count as hardest. Per-condition evaluation quantifies the gain on the hardest regimes (Section 6.4), and the SOH-only / RUL-only models set the ceiling the joint model should exceed.

F.1 RELOCATED MEASUREMENTS

The two results below are stated in condensed form in Sections 6.3 and 6.5; the full measurements are given here.

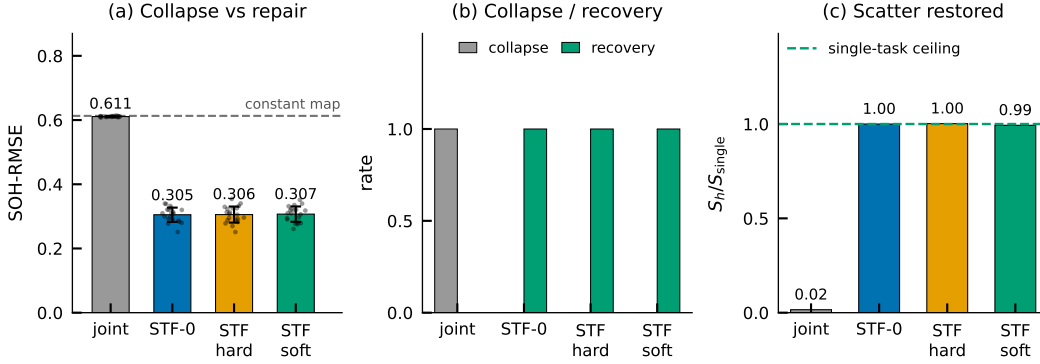


Figure 2: On battery SOH the collapse is visible to the primary metric, and the filter repairs it (unified-protocol battery domain, NASA PCoE, $m=128$, $\alpha=10^4$, 20 seeds per mode). The metric is the log-SOH RMSE of Tables 17 and 15; it is not the SOH-RMSE% of the flagship Table 2, which is a different dataset (MATR) and model, so the two scales are not comparable. (a) log-SOH RMSE (log axis): joint collapses to 0.611, a constant map, while every STF mode recovers to ≈ 0.305 . (b) Collapse rate $1.0 \rightarrow 0.0$, recovery rate $0.0 \rightarrow 1.0$. (c) Relative scatter restored from 0.02 to the single-task ceiling 1.0.

Table 18: Composing the toxicity filter with conflict resolution at supercritical α (5 seeds, three domains). The “combiner” column acts on the unfiltered or filtered branch. On the unfiltered branch no combiner prevents collapse (scatter ≤ 0.15 against the single-task 1.0); the filter alone repairs it, and a conflict operator after the filter changes the metric by less than seed noise. On HAR accuracy stays flat even when collapsed, so only the scatter records recovery.

	combiner	unfiltered (JOINT)			filtered (STF-HARD)		
		sum	PCGrad	CAGrad	sum	PCGrad	CAGrad
Battery	log-SOH RMSE↓	0.611	0.592	0.603	0.294	0.283	0.291
	collapse rate	1.0	1.0	1.0	0.0	0.0	0.0
	relative scatter	0.02	0.03	0.03	1.00	1.05	1.18
HAR	accuracy↑	0.941	0.667	0.952	0.943	0.933	0.940
	collapse rate	1.0	1.0	1.0	0.0	0.0	0.0
	relative scatter	0.15	0.00	0.03	0.97	1.14	1.03
RadioML	accuracy↑	0.293	0.091	0.109	0.301	0.296	0.295
	collapse rate	1.0	1.0	1.0	0.0	0.0	0.0
	relative scatter	0.02	0.00	0.00	0.97	0.99	1.04

Composing a filter with a conflict operator at supercritical α (Table 18) confirms the ordering. (i) Because toxic gradients are aligned, a conflict operator on the *unfiltered* branch has no opposed pair to act on: on battery it leaves the encoder collapsed in every seed (collapse rate 1.0, relative scatter 0.029/0.030 against the single-task 1.0) even as the scalar metric moves — the metric–representation decoupling again. (ii) The filter alone (STF-HARD+sum) drives the collapse rate to 0.0, restores the scatter to 1.00, and recovers SOH-RMSE from 0.611 to 0.294, essentially the single-task ceiling; layering conflict resolution *after* it moves the metric by less than seed noise (0.283/0.291) and leaves the scatter on the healthy branch (1.00–1.18). HAR and RadioML reproduce the ordering: the two interventions complement each other, but only when *sequenced*.

Battery degradation is non-stationary, so BatteryMTL multiplies three *dataset-calibrated* gates over C-rate, temperature, and degradation rate, a soft AND with percentile thresholds rather than fixed physical cuts. It leaves aggregate SOH unchanged but improves the hardest windows: a single-term attribution isolates the fast-charge gain to the rate factor (0.95 SOH-RMSE without it) and the extreme-temperature gain to the temperature factor (2.35 without it). The aggregate rows are in Table 2 (Appendix F).

Table 19: Anti-collapse regularizers at each domain’s Table 15 dose and seeds (ten per cell), added to the plain joint path with weight λ . Each domain block gives the collapse rate over seeds, the primary metric, and the relative scatter (1.0 = the single-task anchor). JOINT, STF-0 and STF-HARD are measured in the same runs, so every row is comparable. Both families pin the representation where the penalty puts it rather than at the anchor.

	HAR		Battery		RadioML	
	coll. / acc.↑	rel. scat.	coll. / RMSE↓	rel. scat.	coll. / acc.↑	rel. scat.
JOINT	1.00 (0.945)	0.15	1.00 (0.610)	0.022	1.00 (0.256)	0.008
STF-0	0.00 (0.945)	1.00	0.00 (0.297)	1.00	0.00 (0.341)	1.00
STF-HARD	0.00 (0.946)	0.99	0.00 (0.299)	1.00	0.00 (0.339)	0.97
<i>VICReg (variance + covariance)</i>						
$\lambda=1$	1.00 (0.950)	0.22	1.00 (0.610)	0.023	1.00 (0.249)	0.007
$\lambda=3$	0.20 (0.946)	0.31	1.00 (0.610)	0.023	1.00 (0.253)	0.007
$\lambda=10$	0.00 (0.944)	0.36	1.00 (0.609)	0.024	1.00 (0.256)	0.008
$\lambda=30$	0.00 (0.926)	0.36	1.00 (0.604)	0.033	1.00 (0.260)	0.008
$\lambda=100$	0.00 (0.902)	0.36	1.00 (0.451)	0.14	1.00 (0.301)	0.011
$\lambda=300$	0.00 (0.873)	0.36	0.00 (0.339)	0.59	1.00 (0.362)	0.035
$\lambda=10^3$	0.00 (0.843)	0.36	0.00 (0.390)	1.95	1.00 (0.320)	0.15
$\lambda=3 \times 10^3$	0.00 (0.831)	0.36	0.00 (0.423)	3.70	0.10 (0.207)	0.45
$\lambda=10^4$	0.00 (0.821)	0.36	0.00 (0.423)	4.54	0.00 (0.215)	1.29
<i>Barlow Twins (redundancy reduction)</i>						
$\lambda=1$	1.00 (0.935)	0.11	1.00 (0.607)	0.11	1.00 (0.367)	0.016
$\lambda=3$	1.00 (0.765)	0.022	1.00 (0.607)	0.13	1.00 (0.330)	0.024
$\lambda=10$	1.00 (0.435)	0.006	1.00 (0.606)	0.15	1.00 (0.121)	0.074
$\lambda=30$	1.00 (0.352)	0.006	1.00 (0.606)	0.16	1.00 (0.105)	0.096
$\lambda=100$	1.00 (0.338)	0.006	1.00 (0.605)	0.16	1.00 (0.126)	0.072
$\lambda=300$	1.00 (0.347)	0.006	1.00 (0.606)	0.16	1.00 (0.141)	0.043
$\lambda=10^3$	1.00 (0.343)	0.006	1.00 (0.606)	0.16	1.00 (0.140)	0.042
$\lambda=3 \times 10^3$	1.00 (0.365)	0.006	1.00 (0.607)	0.16	1.00 (0.148)	0.053
$\lambda=10^4$	1.00 (0.339)	0.006	1.00 (0.607)	0.16	1.00 (0.153)	0.044

F.2 WHY THE PROBE CANNOT BE TRACKED ONLINE

Section 5.1 rejects an online ρ -gate, and the reason is measurable. We re-ran the unfiltered joint path with the monitor on (`--rho-trace`), five seeds per cell, over the full schedule (1730 steps on HAR, 350 on battery). The monitor only records: the aggregate rows of these runs reproduce the BSCALE cells they overlap exactly (28 of 28 matched cells, $\max|\Delta| = 0$ over the metric, the collapse rate, the relative scatter and $\hat{\rho}$), so the trajectories are not a perturbed training run.

Three facts follow. First, the probe starts at the one-batch floor and *rises*: on HAR it runs from 0.017 to 1.9×10^5 at $m=32$, from 0.010 to 1.0×10^5 at $m=64$, and from 0.006 to 3.5×10^4 at $m=256$. Second, the rise is not a toxicity signal. With the toxifier *inert* ($\alpha=0$) it reaches the same magnitudes — 1.9×10^5 , 1.0×10^5 and 3.5×10^4 — and the lowest dose swept ($\alpha=10^{-4}$) lands in the same range, so whatever drives the rise, it is not the auxiliary weight. The only dose-dependence is downward: at $m=32$ the endpoint falls from 1.9×10^5 at $\alpha=0$ to 5.3×10^2 at $\alpha=10^{-2}$ and 1.1×10^2 at $\alpha=3 \times 10^{-2}$. Third, because the probe rises, the gate $\alpha\hat{\rho} \geq 1$ opens at *every* dose eventually: in all five seeds of ten HAR cells whose endpoint scatter is at or above the crossing (from 0.50 to 1.04) it opens between steps 110 and 590 of 1730. An online gate built on the running probe would therefore delete representation directions on runs that never collapse.

Battery is the contrast that motivates calibrating per domain rather than tracking: there the probe stays $O(1)$ (0.29–0.49 at step 0, 0.3–2.9 at the end), and the gate opens in one of the thirty-five seeds whose scatter stays above the crossing, at the final step. Whether a fixed online rule is inert or spurious is set by the domain’s primary-gradient scale, which is what the healthy-branch calibration measures once. `repro/scripts/rho_trace_analyze.py` prints both tables from the shipped runs.

F.3 WHAT THE CALIBRATION COMMITS TO: THE READ TIME SETS THE GATE’S FLOOR

The calibration reads $\hat{\rho}$ once, on a detached healthy anchor, and both the gate and the rank follow from that single read; every shipped run reads it after four primary-only epochs. To see what that choice is worth we re-ran the identical calibration with *only* the read time W varied over $\{1, 2, 4, 8, 16, 32\}$ — HAR and battery at five widths 32–512 and ten seeds, NYUv2 at four widths and ten seeds. The $W=4$ column is the shipped configuration and reproduces it exactly: $\max |\Delta| = 0$ across all 100 HAR and battery cells, and $\hat{\rho} = 2.258/2.008/1.637/1.285$ on NYUv2 against the 2.26/2.01/1.64/1.29 of Appendix E. The sweep therefore varies nothing but the read time.

The read is a trajectory quantity, and how strongly it moves is a property of the domain. On HAR the probe at $m=32$ climbs from 0.63 to 6.9×10^4 across the sweep — five decades — and at $m=512$ from 51.9 to 1.6×10^4 ; on battery it stays $O(1)$ throughout (0.96 to 1.92 at $m=32$, 0.058 to 0.67 at $m=512$); on NYUv2 it moves by a factor of 2.2 (1.94 to 4.19 at $m=32$). The exponent $\alpha_c = 1/\hat{\rho}$ inherits the drift: read as a width-scaling, the probe exponent runs from -2.00 to $+0.55$ on HAR — it changes sign — from $+0.37$ to $+0.97$ on battery, and from $+0.11$ to $+0.27$ on NYUv2. This is the measured reason the paper reads the capacity law off the crossing for HAR, battery and RadioML and off the probe for NYUv2 (Section 7.1, Appendix E): a probe exponent is a property of the domain only where $\hat{\rho}$ is stable in the read time, and where it is not, the probe’s width-scaling is dominated by how fast each width’s primary gradient decays — the denominator of $\hat{\rho}$ — rather than by the encoder’s capacity.

The same mechanism fixes what the gate can see. The gate opens on $\alpha\hat{\rho} \geq 1$, so the read time sets its floor $1/\hat{\rho}$: on HAR at $m=64$ the floor is 1.34 at $W=1$, 0.181 at $W=2$, 9.07×10^{-3} at $W=4$ and 1.7×10^{-5} at $W=32$, against a measured transition at $\alpha_c \in [0.001, 0.01]$; only $W \geq 4$ puts the floor at or below the dose the filter is meant to defend. Measured on the shipped HAR repair recipe ($m=64$, ten seeds, sixty epochs, JOINT/STF-0/STFCAL in the same run) at a marginal dose $\alpha = 0.03$, three times the transition: at $W=1$ and $W=2$ the gate stays shut in all ten seeds and STFCAL collapses in 10/10, with metric 0.947 and relative scatter 0.19 and 0.20 — the unfiltered joint numbers (0.949, 0.19) to within seed noise — whereas from $W=4$ on the gate opens in every seed, STFCAL collapses in 0/10 and the relative scatter returns to 0.99–1.00. At a dose far above any floor ($\alpha = 100$) the gate opens for every read and the repair is independent of the read time (0/10 collapsed at all six). A warm-up that is too short therefore does not damage the run: it disables the filter, visibly (the gate is reported shut) and at no cost (the run is the unfiltered run). The operating condition is checkable before deployment — choose W so that $1/\hat{\rho}$ lies at or below the dose to be defended against — and at four epochs it is satisfied, the floor landing on the measured transition. `repro/scripts/warmup_sensitivity.py` prints both tables from the shipped runs.

G ANTI-COLLAPSE REGULARIZERS AT MATCHED DOSES

Table 19 runs the variance/covariance family of Table 16 at each domain’s Table 15 dose and seeds, adding the penalty to the plain joint path with weight λ on the shared representation: VICReg’s variance hinge and off-diagonal covariance term, and Barlow Twins’ redundancy reduction. VICReg’s invariance term is dropped, there being no augmented view to be invariant to. Every row is compared against the same JOINT run, so the baseline is the unfiltered path and not a re-tuned primary objective.

Neither family restores the representation to the single-task anchor, and the weight needed to reach the criterion is a property of the domain, not of the penalty: VICReg needs $\lambda=10$ on HAR, $\lambda=300$ on battery and $\lambda=10000$ on RadioML — three orders of magnitude apart, with the collapse rate still 1.0 at $\lambda=1, 100$ and 1000 respectively — and Barlow Twins never clears it in any of the three. Above that weight the representation settles where the penalty puts it. On HAR the variance hinge saturates: from $\lambda=10$ to 10^4 the relative scatter is flat at 0.36 while accuracy falls from 0.944 to 0.821, and the accuracy at the clearing weight is indistinguishable from both JOINT (0.945) and STF-0 — the HAR metric is collapse-insensitive, as Table 17 reports. On battery the scatter is monotone in λ and crosses the anchor, and the primary metric has an interior minimum near the crossing: 0.339 at $\lambda=300$ (relative scatter 0.59) against the 0.297 that STF-0 reaches at the anchor, degrading on both sides (0.390 at 10^3 , 0.423 at 10^4). STF-0 is better on 9 of the 10 seeds. RadioML separates the criterion from the metric outright: at $\lambda=10000$ the collapse rate does reach 0.0, but

the relative scatter has overshot the anchor to 1.29 and accuracy is 0.215 — below the 0.256 of the *collapsed* JOINT path and far below the 0.341 that STF-0 reaches at the anchor, so the penalty has bought a healthy scatter at the cost of the task it was added to help. Barlow Twins drives the absolute scatter to 0.011 on HAR, 0.012 on battery and 0.013 on RadioML, 172, 6 and 24 times below the respective anchors: it clears the criterion nowhere, and on HAR it takes accuracy down to 0.34–0.44, on RadioML to 0.11.

A finer grid around each crossing ($\lambda \in \{2, 5, 20, 50, 200, 500\}$ on HAR and $\lambda \in \{200, 500, 700, 1500, 2000, 5000\}$ on battery, ten seeds) narrows both readings without changing them: the battery threshold sits between 150 and 200, and the lowest battery metric anywhere on the grid is 0.339 at $\lambda=300$ (relative scatter 0.59), still above STF-0; the HAR scatter is flat at 0.36 for every weight above the crossing. The RadioML crossing was not refined, so its threshold is only located between $\lambda=1000$, where the collapse rate is still 1.0, and $\lambda=10000$, where it reaches 0.0.

The collapse criterion is a threshold on relative scatter. A penalty with a global weight has a fixed point of its own, set by λ rather than by the data, and the single-task anchor is not where it lies.